

Artificial Intelligence and Data-Driven Innovation in the Automotive Industry: A Systematic Literature Review

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1 Introduction

The fast evolution of Artificial Intelligence (AI), Machine Learning (ML) and data driven technologies is transforming deeply every aspect of the economy and industry. Recent reports highlight that the automotive industry will experience a growth in the next decade given by the technological development and enhancement of connected vehicles, predictive maintenance, decision support system and autonomous driving. Market analysis forecast that the global automotive AI market could have a growth rate of 23% or over (GVR Next Generation Technologies Research Team, 2024). To follow the trend enterprises like Stellantis are starting to collaborate with big tech companies in order to accelerate the development (Microsoft, 2026).

In the present study is analyzed how this technology is reshaping the automotive industry, not only the vehicle functionality but also the broader configuration of product, services and the entire business model.

Despite the growing adoption of AI and data-driven solutions in the automotive industry, the managerial literature and academic remains fragmented and spanned through multiple disciplines such as engineering, information system, strategic management and operation management. Several papers highlight this research gap, starting from evidencing the absence of a complete frameworks supporting data-driven decision making and organizational integration (Vacarescu et al., 2025), passing through the evidence of a reduced studying of the predictive analysis and their strategical application in the sector (Ranabhatt et al., 2025), arriving to the emphasize of scarcity of AI or ML application in context like co-creation, integration to achieve ESG objectives and UX dimension (Ghantous & Alnawas, 2021) (Li et al., 2021) (Cescon & Peruzzini, 2026).

To address this gap and identify the main technological approaches, domain of application and innovation strategy; a structured and systematic analysis of the existing knowledge is needed. This study aims to provide a comprehensive and data-driven overview of the actual literature material related to the topic.

Specifically, this thesis addresses the following Research Question:

RQ1: What are the main AI, Machine Learning and data-driven approaches adopted to innovate the automotive offering and business model?

To answer this question a systematic literature review was conducted combining traditional qualitative assessment with quantitative text-mining techniques. The study follows the PRISMA methodological framework. The workflow of the thesis involved the collection, screening and validation of scientific papers from major academic databases ensuring methodological rigor and transparency through all the process. The study was supported by the implementation of structured filtering criteria and systematic evaluation of relevance, methodological quality and consistency.

After the identification of useful studies, the research adopted an analytical framework that can capture patterns and thematic structures in literature. LDA Topic modelling techniques were applied to validate the sample and identify the main recurring themes, transforming a huge amount of unstructured textual data into interpretable thematic categories, facilitating a deeper comprehension of the analyzed theme. The analytical process was enabled by KNIME Analytical Platform.

The results of the analysis were interpreted through graphical representation and descriptive statistics to obtain the identification of the main technological domains, strategic application areas and evolving research trends. Specifically, the thesis evidences the growing relevance of the AI use in digital transformation initiatives, predictive analytics, intelligent decision support systems, costumer centric innovation strategies and autonomous driving. All the findings contribute to clarifying how technological and managerial mechanisms are influenced and enhanced by artificial intelligence and technologies. Topic modeling results also evidence that the role of AI-enabled system is redefining all the organizational processes, competitive dynamics and customer interaction models through all the automotive value chain.

Looking at the previously cited findings, this thesis also provides possible research agenda area to enhance the understanding of AI-driven technologies in relation to sustainability objectives, business model transformation and user-centered innovation.

From a theoretical perspective, the study contributes to the actual academic environment by exposing an integrated evidence base synthesis of the fragmented literature domains, combining organizational, managerial, technological, and sustainability topics within a unified analytical framework. The research also contributes methodologically by adding

a systematic literature review procedure within LDA topic modelling techniques to support the emerging evidence in the related fields.

From a managerial perspective, the study provides practical insights into the automotive industry and to the companies operating in the market explaining how the increasing digitalization changes the environment. The outcomes help the managers in understanding how AI and data-driven technologies can enhance strategic decision making, optimize performance, improve customer engagement and perception, support sustainable objectives and consolidate or create competitive advantage. Moreover, this thesis gives guidance for practitioners seeking to understand how invest to follow the evolving market demands, ESG objectives, digital transformation and obtain a future development in the automotive industry.

2 Research gap

Previous literature has increasingly studied the use of AI, machine learning and data-drive approach in the automotive industry. However, the investigation contributes to achieving results and findings on the subjects but leaving a fragmented framework across isolated domains such as user experience, sustainability, management and demand forecasting. Different papers evidence the lack of an integrated perspectives that can obtain a complete view on the relationship between technological innovation and business model transformation or and value creation process.

Specifically, recent study emphasizes the absence of the same (Vacarescu et al., 2025). Similarly, research on demand forecasting in automotive sector identified the need for further and broader analysis in combining strategic and operational limitation with predictive analytics (Ranabhatt et al., 2025). Another paper highlights the limited understanding of how digital technologies in the sector can achieve co-creation and customer integration (Ghantous & Alnawas, 2021).

Looking at the user experience, costumer centered approach and sustainability additional gaps emerge. Present studies evidence the necessity of more researches on AI, ML and data-driven approaches development in topics like ESG objectives and sustainable product (Li et al., 2021) (Carvalho et al., 2024). Other recent reviews on user experience and human machine interaction underline the absence of holistic approaches that able the full linking between UX design, design, usability and data-driven approach (Cescon & Peruzzini, 2026) (Ebel et al., 2021).

To address these gaps the present research focuses on analyzing the main AI, ML and data-driven approaches used to innovate the automotive offering and business model through a SRL combined with topic modeling analysis. The study gives an integrated overview of technological, sustainability, user experience and organizational dimension in the evolving environment of the automotive industry.

3 Methodology

To understand how the actual automotive industry is using AI and machine learning methods the research is based on different steps, where each one has a specific and

fundamental function. Every phase can be applied, with some changes, in other research areas by downloading the knime workflow that follows ([knime workflow](#)) (also visible on knime hub by searching alessandrocaros profile and looking for the thesis folder, thesis 3 file's).

3.1 Research design

This study applies a multistage methodological framework to ensure both rigor and scalability. It combines a systematic literature review (SRL), following the guidelines proposed (Moher et al., 2009), and a Latent Dirichlet Allocation (LDA) topic modeling, following the guidelines proposed (Blei, 2012). Using this methodology the selection of high quality and relevant academic contribution is ensured by SRL and the data-driven identification of the underlying thematics is allowed with the LDA. To complete the preliminary structure of the research the following research question was formulated; RQ1: *What are the main AI, Machine Learning and data-driven approaches adopted to innovate the automotive offering and business model?*

3.2 SRL protocol

In this section there is a description of how the systematic literature review was done, starting from the research of the literature, arriving at the extraction of the final dataset that will be used in the text mining analysis.

3.2.1 Search strategy

All the material was collected using a strict review protocol and through recognized search engines. Specifically, Scopus, ScienceDirect, IEEE Explorer and Google Scholar were used. The major academic databases were targeted using a search string that integrates all the core concepts of the research using logical operators: [AI/machine learning] AND [automotive industry/vehicle manufacturing] AND [Business Innovation/business model]. In all the search engines the words searched and the logical or filtering parameters were the same.

#Ex. Scopus query => TITLE-ABS-KEY (("automotive industry" OR "automobile industry" OR "car industry" OR "passenger vehicle industry" OR "passenger cars" OR "vehicle market" OR "car market" OR "automotive sector" OR "auto industry" OR "car manufacturing industry" OR "car retail" OR "car dealership") AND NOT ("public transport" OR "public transportation" OR "bus" OR "buses" OR "metro" OR "subway" OR "tram" OR "railway" OR "rail" OR "transport" OR "train")) AND TITLE-ABS-KEY ("artificial intelligence" OR "AI" OR "IA" OR "machine learning" OR "ML" OR "data-driven" OR "data analytics" OR "big data" OR "advanced analytics" OR "predictive analytics" OR "data science" OR "algorithmic decision-making") AND TITLE-ABS-KEY ("business model" OR "business model innovation" OR "business model transformation" OR "value creation" OR "value capture" OR "value proposition" OR "revenue model" OR "monetization" OR "platform business" OR "model ecosystem" OR "servitization" OR "digital transformation" OR "subscription model" OR "pay per use" OR "MaaS" OR "product innovation" OR "service innovation" OR "digital offering" OR "smart services" OR "personalized offering" OR "mass customization" OR "product-service system" OR "PSS" OR "solution offering" OR "customer-centric offering" OR "pricing strategy" OR "dynamic pricing" OR "price optimization" OR "customer analytics" OR "customer segmentation" OR "churn prediction" OR "customer lifetime value" OR "CRM" OR "personalization" OR "recommendation" OR "digital retail" OR "online sales" OR "e-commerce" OR "omnichannel" OR "customer journey" OR "demand forecasting") AND PUBYEAR > 2019 AND PUBYEAR < 2027

3.2.2 Screening & selection process

The research resulted in a broad set of publications, which included different types of text and papers. From an initial data set composed of 622 documents the final study focuses on 78 papers. All the exclusions, from the query and after, are done using the following criteria to ensure relevance and specificity:

- All the documents that are antecedent respect to the 2020 were removed from the sample as they are considered outdated (n.=95).
- Not relevant articles were removed (n.=46) by specifying in the query with the “AND NOT” logical operator that there must not be any words corresponding to the following set: "public transport" OR "public transportation" OR "bus" OR "buses" OR "metro" OR "subway" OR "tram" OR "railway" OR "rail" OR "transport" OR "train".
- All the text that are not correspondent to an article type were removed (n.=348).
- All the duplicate records were removed (n.=3).
- All the papers that came from not reliable journals were excluded from the sample (n.=38), as criteria of exclusion were used the Quartile of the journals of belonging. Specifically, if a paper came from a journal that has a Q lower than 2 were excluded.

At the end of the selection process the database was composed of 78 papers, the 12,54% of the initial records.

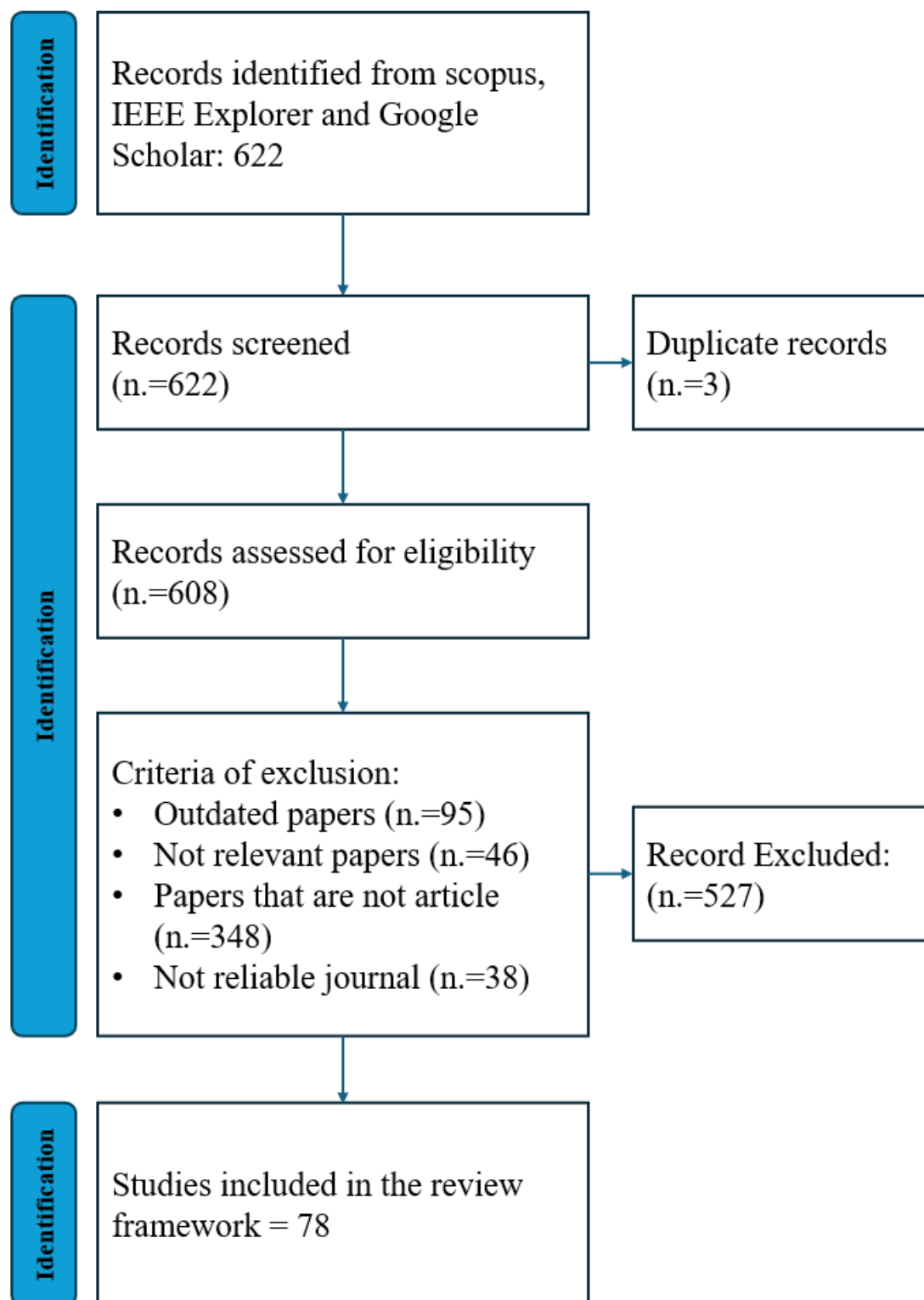


Figure 1 Prisma Workflow

Overview of the reviewed framework:

Paper per years

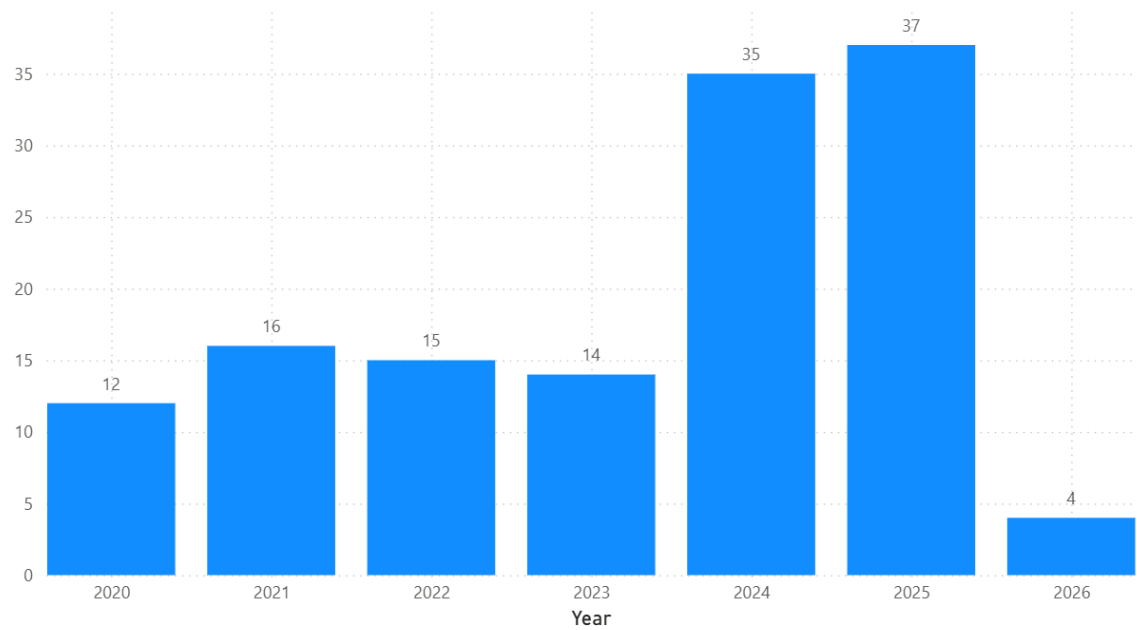


Figure 2 Paper per year

Obviously, the number of papers in 2026 are not representative. By using the estimating function of excel is conceivable that the papers in 2026 will be between a range of 29 and 54.

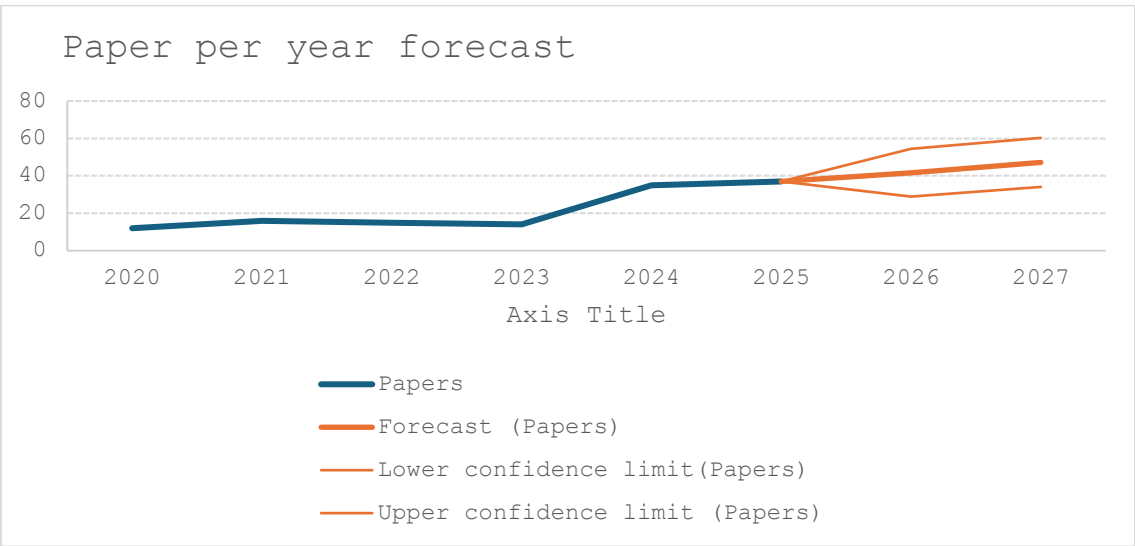


Figure 3 Paper per year forecast

A table that summary for each paper the authors and years, the methods used in the study, the context, the focus or the topic and the key findings with each description are provided (11.6).

3.2.3 Data extraction and data set construction

The different excel files obtained from Scopus/Science Direct, IEEE Explorer were imported in KNIME Analytics Platform. The complete Google Scholar excel file where more difficult to obtain; in order to import all the bibliometric information, the automatic citations csv exportable by Scholar must be integrated with abstract, type of text, DOI and Year. To do this implementation automatically a Python script that query the Semantic Scholar's API was used (Appendix paragraph (11.1)). This open API is a means that can be freely used for search academic paper info with some limitations. All the mechanisms have been implemented in the KNIME workflow (all the code is explained with the within comments in the KNIME node Python snippet, for example if after a request the server respond with a 200 code it means that the query has been successfully completed, or 403 forbidden or 404 not founded). For the papers or the information that cannot be found with the Semantic Scholar API a manual integration has been done (the script is programmed to highlight which records have been integrated and which not so hybrid approach can be used).

At this point, with all the excel files imported and completed, it is possible to define a standardize form of DB to make all the dataset confrontable and uniform to perform the last screening operations and, after them, the LDA algorithm. The final sample will have the following column structure: "Authors", "Title", "Year", "DOI", "Abstract", "Author Keywords", "Document Type", "Source title", "Publisher", "Article Citation Count", "Method", "Context", "Focus / Topic" and "Key Findings".

Renaming all the column from the different search engines in the same way make able to exclude the duplicate records by just looking at the DOI code and removing the not according type of document from the research.

3.2.4 LDA preprocessing

To achieve a rigorous and robust analysis of all the abstracts a pre-processing phase must be done. In the knime workflow all the preprocessing steps are in the "Data Preparation

for topic models” box, where there are different meta nodes (a knime structure in which a group of nodes are included). There are few main actions that must be done:

- Transform the text of the abstracts into a file type that the following Knime nodes can read and analyze. To perform this action the “String to document” node has been used.
- After obtaining the right format is possible to perform multiple operations to simplify the text. Here the mains: The first node used is the “stop word filter” that eliminates the frequent word that, given their enormous popularity (Zipf’s Law [ERROR! REFERENCE SOURCE NOT FOUND.](#)), would not be representative of the sample. The second is applying the Kuhlén Stemmer, an algorithm that takes all the words to their root (ex. Given => Give).
- Also, a general simplification of the text has been done by removing the upper-case charts, number, punctuation and all the parts of the discourse that cannot be captured by the LDA.
- Lastly, the words that are not distinctive like AI, automotive, machine and with those also the extremely infrequent were removed.
- After the removal of all the possible problems given by the vocabulary the documents that have less than 10 words were removed (n.=0).

3.2.5 LDA setting

The last step to perform an optimal topic modeling analysis is to identify the right hyperparameters: the number of topic (k), the α and the β .

To understand the right k the two following indexes were analyzed.

1. Perplexity index: used to evaluate how good the predictive performance of a topic model is on unseen data. Formally, it is defined as the inverse geometric mean of the per-word likelihood of a held-out test set. The lower the perplexity index is, the better is the model fits on the data distribution.

In the model of the study after running the perplexity index with a series of k between 1 and 15 was find out that the elbow point coincides with an optimal number of topics equals to 9 .

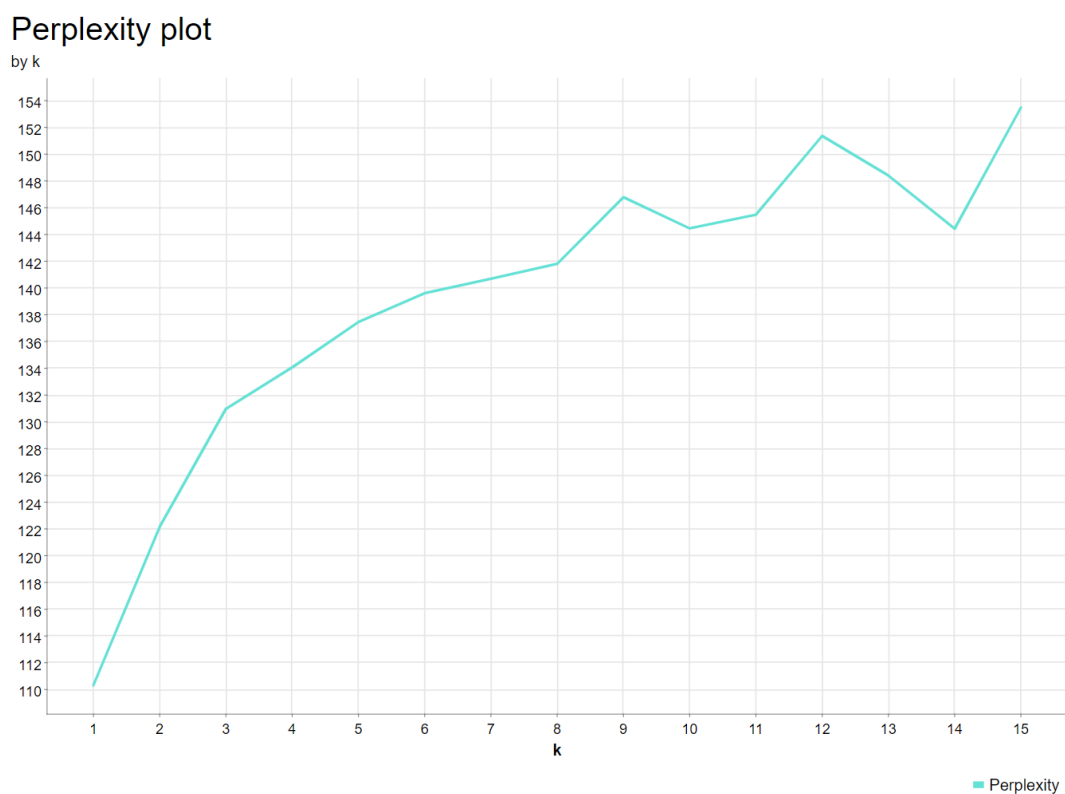


Figure 4 Perplexity plot

2. Chi square index: used to evaluate the goodness of fit and the statistical independence between the identified topic and the document text. The statistical principle is to test the null hypothesis that the observed topic distributions are independent respect to the document categories. The higher the chi square index is, the higher is the distinctiveness between the topics. As is detectable from the graph, after running the CHI-Square within a range of k between 1 and 15, the elbow point coincides with an optimal number of topics equals to 9.

CHI-Square plot

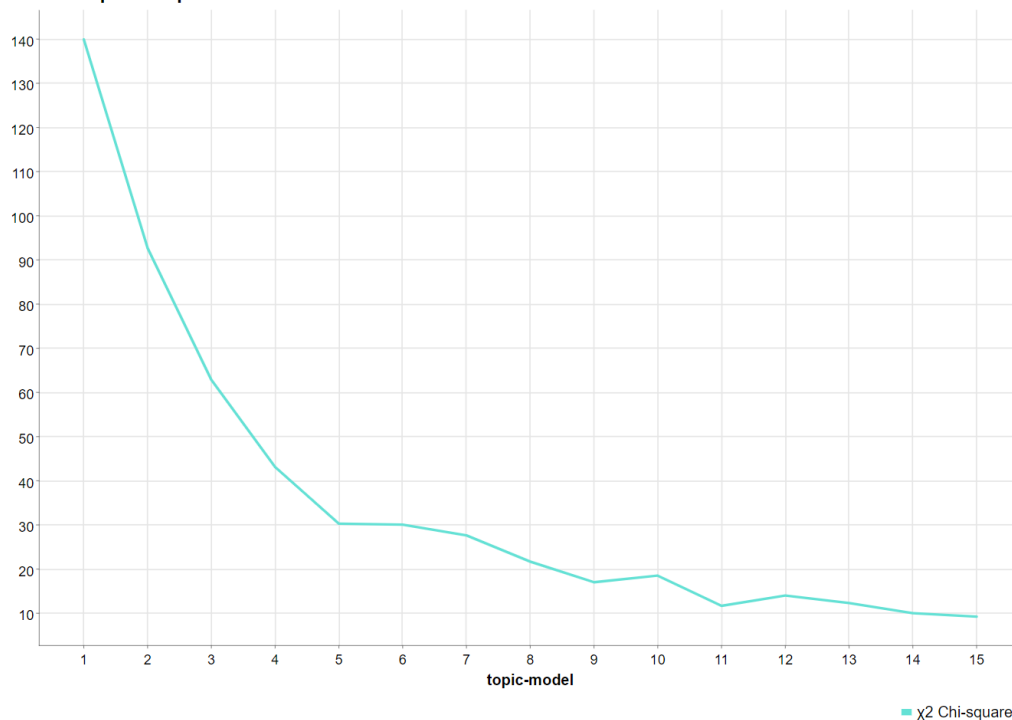


Figure 5 CHI-Square plot

The last two variables were settled on the described logics:

1. Alpha (α): represents the concentration of document parameters. If alpha is a low value (less or equal than 0,1) it means that the documents are described from a low number of topics. In this specific model the alpha was settled on 0,1.
2. Beta (β): represents the concentration of topic parameters. If beta is a low value (less or equal than 0,01) it means that the topics are described from a low and determinant number of words. In this specific model the beta was settled on 0,01.

After that all the preprocessing and setting action has been done is possible to run the LDA and obtaining consistent, valid and rigorous results.

3.2.6 LDA results

The LDA algorithm has defined the words contained in the topics and, for each of the 78 abstracts, the assigned topic.

Specifically, the following words have been identified for each topic:

Topic Id	Terms	Label
topic_0	drive, battery, fault, energy, condition, truck, logistic, cell, probability, fuel	Battery Fault Diagnostics
topic_1	innovation, integration, manufacture, sustainable, challenges, sector, role, process, company, opportunity	Industrial Innovation Management
topic_2	service, machine, demand, propose, forecast, demand-forecast, cost, time, customer, operation	Demand Forecasting Operations
topic_3	policy, knowledge, information, electric, data-driven, innovation, energy, connect, value-creation, dynamic	Data-Driven Mobility Governance
topic_4	product, sustainability, customer, charge, consumer, manufacturer, user, features, analytic, platform	Sustainable Product Design
topic_5	safety, provide, standard, software, framework, engineere, algorithm, challenges, future, dnn	Safety Systems
topic_6	user, experience, perceive, control, effect, augmentation, interaction, content, conduct, enrich	User Experience Design
topic_7	fuzzy, neural-network, material, decision-make, product, use, barrier, variables, support, level	Decision Support
topic_8	company, practices, digital, value, transformation, importance, technological, supp-chain, firm, digital-transformation	Digital Transformation Strategy

Those are the distribution (and the relative one) for the assigned topic to the papers:

Label	Absolute count	Relative count
Battery Fault Diagnostics	4	0,051
Industrial Innovation Management	11	0,141
Demand Forecasting Operations	12	0,154
Electric Mobility Policy	10	0,128
Sustainable Product Design	6	0,077
AI Safety Systems	9	0,115
User Experience Design	2	0,026
AI Decision Support	9	0,115
Digital Transformation Strategy	15	0,192

Absolute count per topic

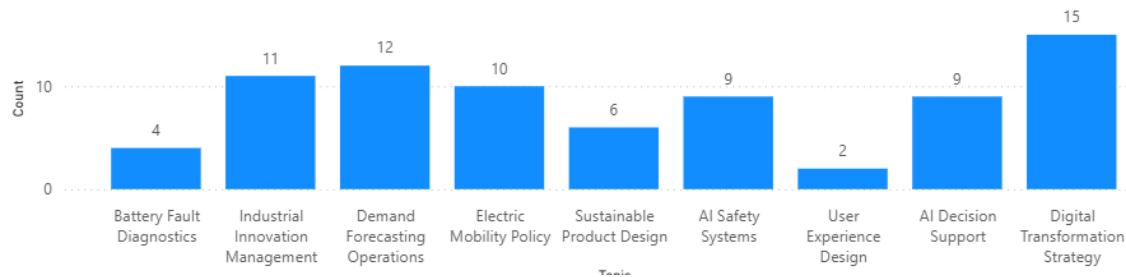


Figure 8 Relative count per Topic

Relative count per topic

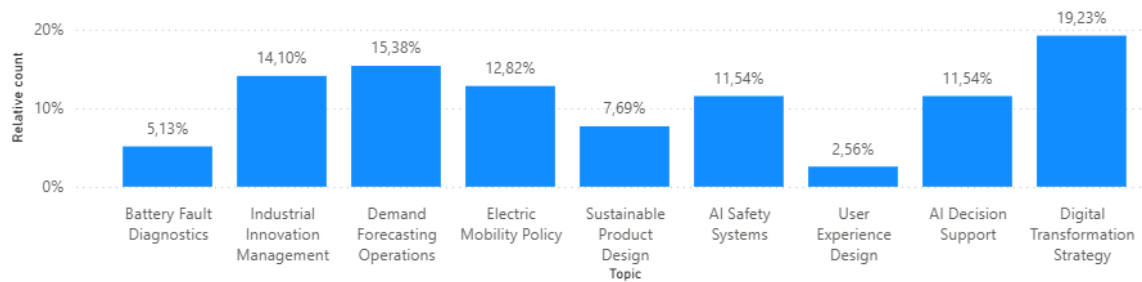


Figure 7 Absolute count per Topic

Words weight colour coded for Topic

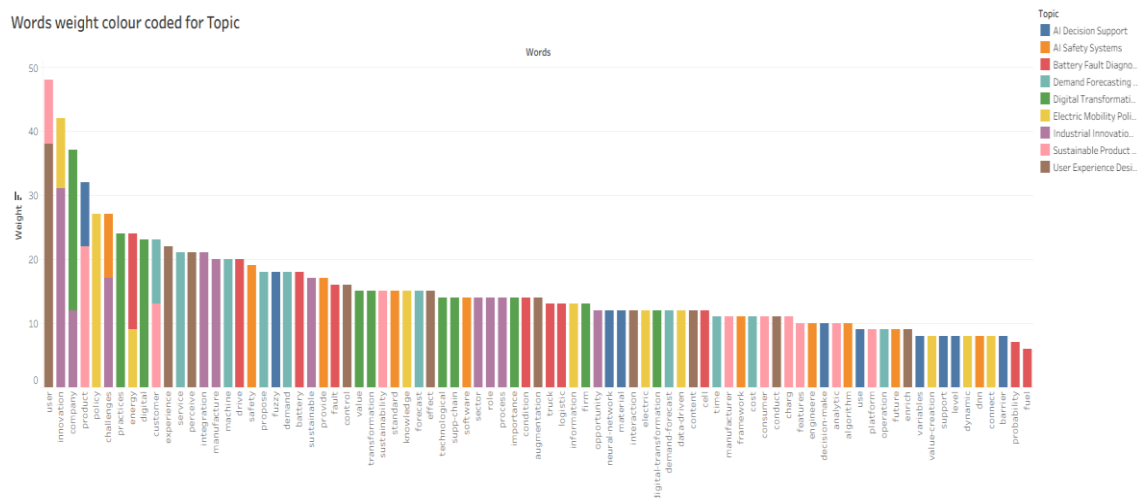


Figure 6 Words weight color coded

By looking at the words that compose the topic and the distribution of the topic assigned is affirmable that the LDA algorithm is consistent and obtains valid, rigorous and analyzable results.

3.2.7 LDA visualization

To visualize the LDA algorithm results, two graphs have been done.

1) Words clouds:

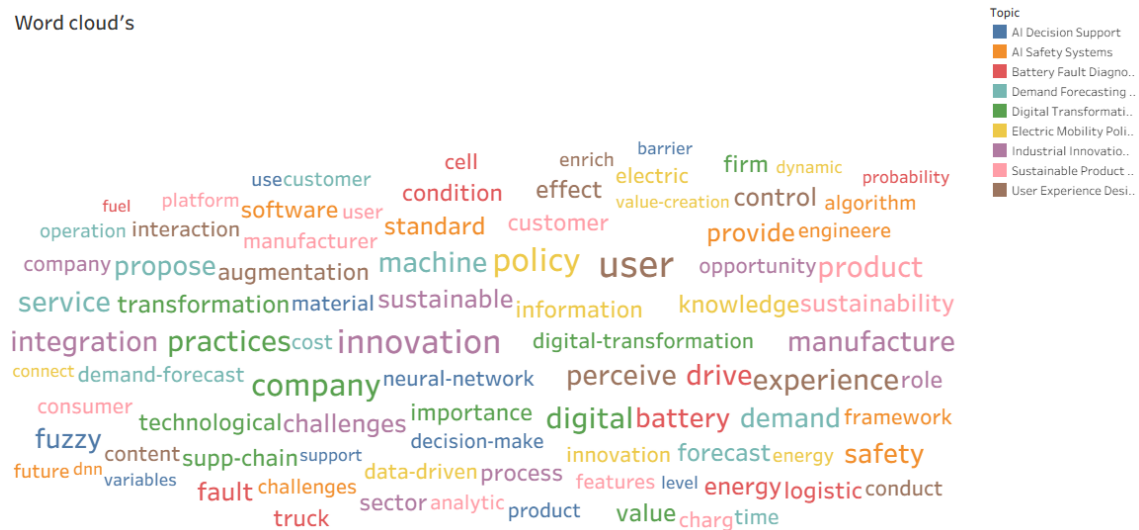


Figure 9 Words cloud's

Each word is color coded based on the topic of appurtenance.

2) LDA visualization: to achieve this representation several works have been performed, starting with the extrapolation of the data from the Topic extraction (Parallel LDA) node in knime, from which 3 outputs are provided. One where there is the document collection with topic assignments and the probability for each document to belong to a certain topic, the second one in which there is topic models with each term and their weight per topic and a last one with a table with statistics for each iteration (like the log likelihood use to calculate the Perplexity index above mentioned).

After the extrapolation of the information and their adaptation through pivoting node and math formula node a conversion in Json file and exportation is necessary (the exportation is needed due to a temporary technical problem but the py script also is implemented in knime in case of resolution).

With the help of Visual Studio Code a merge with python language has been done between the four Json files created previously, obtaining one Json file containing 4 information:

- For each word the weight that has in the single topic (ϕ (φ))
- For each topic all the weight that compose single abstract (θ (θ))
- The length of each document analyzed
- The absolute frequency of each term that composes the topics.

By re-importing the merged Json file in knime trough Json reader node and by applying a python script node in which are encoded the relationships between the data and the dimension of the graph the LDA visualization has been achieved. (Link to access to the html viz <https://alessandro391.github.io/tesi-lda/#topic=0&lambda=1&term=> or QR code for the same:

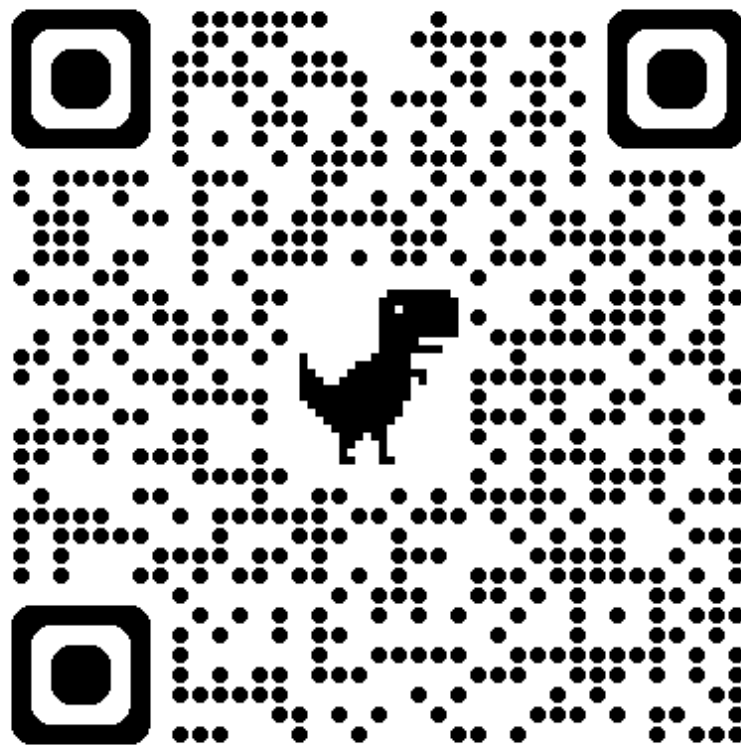


Figure 10 pyLDAviz QR Code

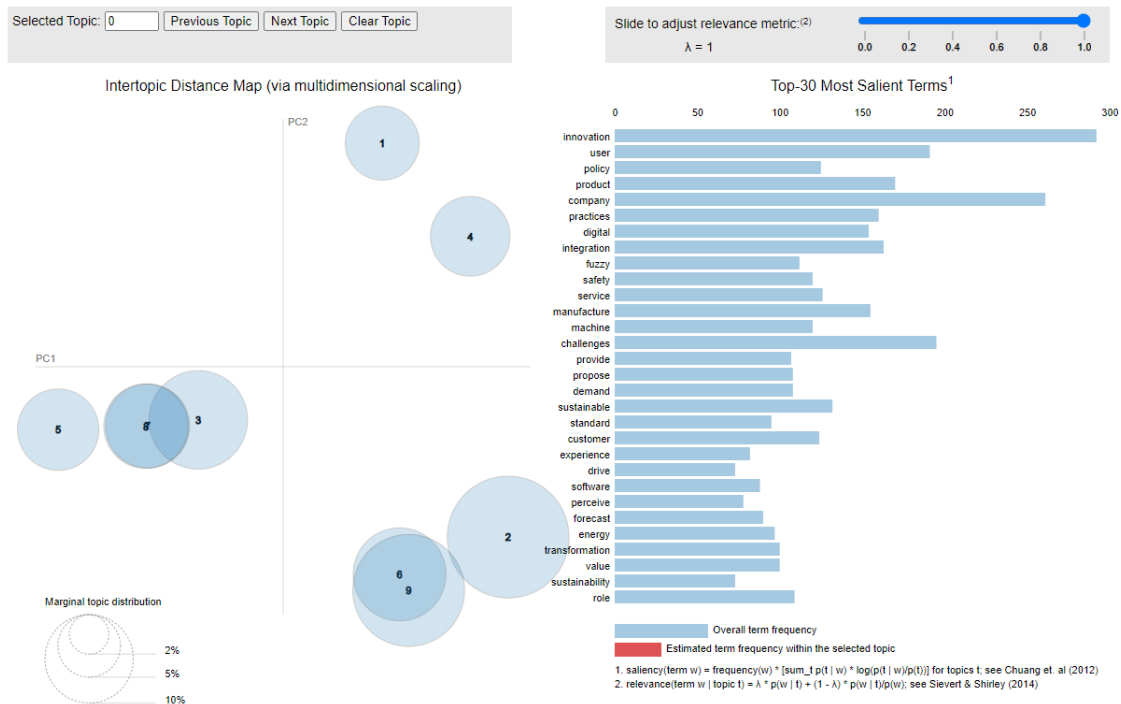


Figure 11 pyLDAviz example

The interpretation of the visualization is the following:

- On the left there is a bidimensional graph where the x axis is represented by the PC1 (Principal component n.1) and the y from PC2; those components don't have a strict sense but express the differentiation between the topics that are represented by the bubble, the bigger the bubbles are the more is important the topic in all the corpus of the analyzed text. All the bubbles are selectable and interactive with the other representation. If the topics are overlapping this means that they contain similar words but with different meanings.
- On the right a composed bar chart has shown, the blue bar represents the overall term frequency in the all the text. By clicking on one of the topics mentioned before, the red bars are also visualized and them express the estimated term frequency within the selected topic.
- The default order of the bars is based on the salience of each word; this characteristic is defined as how much a word is useful to distinguish between the topics. The salience is mathematically defined as:

$$saliency(w) = freq(w) * \sum_{t=0}^n p(t|w) * \log\left(\frac{p(t|w)}{p(t)}\right)$$

- The parameters lambda (λ) influences relevance and, as consequence, the order with which the words are shown when a topic is selected. Selecting a higher λ the most common words are placed in the top, but those words, probably, have a more generic sense, so are less explicative of the selected topic; a lower λ evidence the words that are probably less present but more representative.

To explain this effect the formula of word's relevance must be analyzed:

$$relevance(w|t) = \lambda * p(w|t) + (1 - \lambda) * \frac{p(w|t)}{p(w)}$$

By putting the λ equals to 1 the relevance of the word is based only on the probability of a word to be in that topic, that, obviously, is higher if the word has a higher frequency. Vice versa, imposing the λ equal to a lower value gives more importance to words that are only in that specific topic, this by comparing the word frequency in the topic on the word general frequency in all the sample.

With all the tables and the visualization, the chance to interpretate and understand the full flow of the data and their outcome has been achieved.

3.2.7.1 Interactive dashboard

Also, an interactive power BI dashboard was structured. (https://luiss-my.sharepoint.com/:u:/g/personal/alessandro_carosi_studenti_luiss_it/IQDAkSUCnZC9Sq05hggikMMrAQGvTBQ3827-9pbLD3AnMos?e=QJ4hiY). The dash can give a rapid and complete view of topic frequency and the words weights. To obtain the interactivity in all the graphics a relation model are built, using the slicer (filter) is possible to choose which topics analyze and in the horizontal bar graph the selected argument will be automatically highlighted in orange, at the same time, the vertical bar graph will be sorted and filtered evidencing the word in the selected topic. Power BI will automatically exclude the topics that are not selected in the filter so to maintain them the DAX language was used to force the correct and clear relations.

The same program language was use in the apposite section to apply also the different color to the bars in the horizontal bar graph.

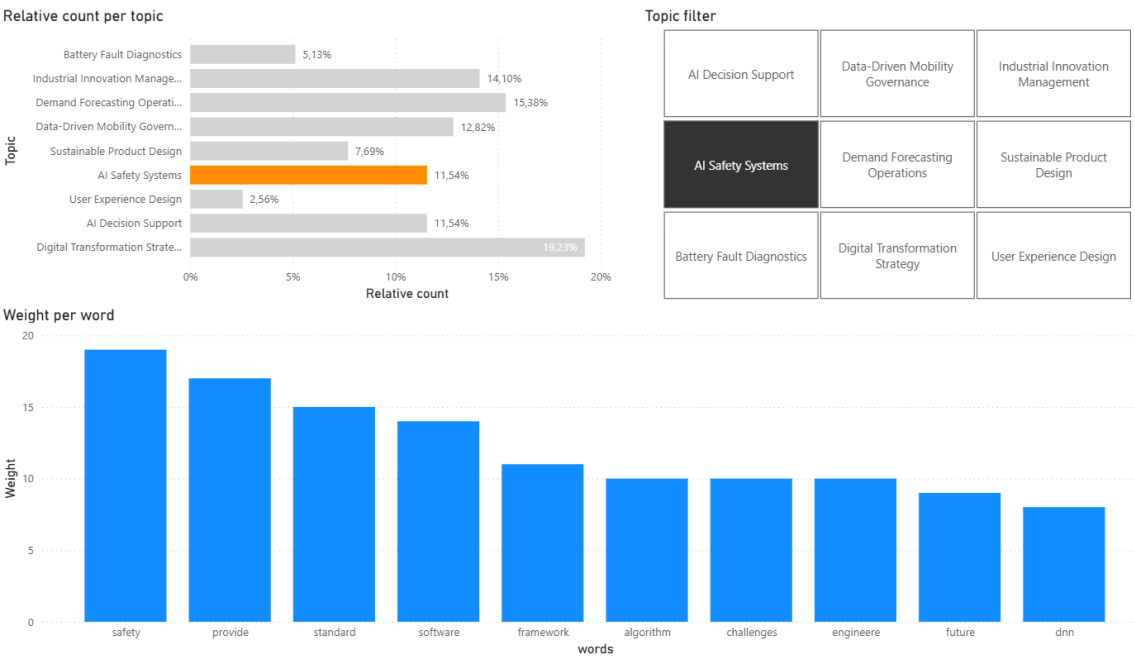


Figure 12 Interactive Dashboard

4 Results

In this section the general analytical results were analyzed.

4.1 Overview and topic distribution

The topic modeling analysis was done on a final dataset of 78 reviewed publications related to the use of AI, ML and data-driven innovation in the automotive industry. The Latent Dirichlet Allocation (LDA) algorithm was settled on 9 topics (decision based on the perplexity index and the CHI-square index) in order to provide nine distinct research topics and a structured representation of the main thematic argument present in the literature. The distribution of the topics is balanced, without a single topic dominating.

4.1.1 Battery Fault Diagnostics (topic_0)

The topic that regards Battery Fault Diagnostics (5,1%) indicating emerging research streams related to product innovation and predictive maintenance. Specifically, using clustering, dimensionality reduction and other statistical models an extremely high identification of problems that cause battery fault and performance driver is achieved (Jiangwei et al., 2024). The implementation of deep neural networks enables the discovery of hidden reasons and relationships between high numbers of variables or situations and the outcomes of the electrical product (T. Wang et al., 2025). The high digitalization and use of technology like blockchain can enhance the value of the supply chain but reveal several problems related to data quality and cyber security evidencing the need of robust and safe infrastructure (Markov & Vitliemov, 2020).

4.1.2 Industrial Innovation Management (topic_1)

Industrial Innovation Management topic (14,1%) represents the strategic dimension of tech development and adoption during the process, emphasizing the role of innovation management and organizational capabilities in enabling digital transformation. The use of AI and machine learning improves fuel economy, reduces emissions and enables advanced fault diagnosis (however an important gap remains because the effective applications in the sustainability environment are marginal and not highly impactful) (Slouma et al., 2026). Smart manufacturing outperforms classic systems but the implementation of the first relay on investment and capability to manage technological

complexity (Kamble et al., 2020). Similarly, firm performance improvement is based on planning and external collaboration in combination with sustainability and strategic information within the supply chain (Piepoli et al., 2024). AI significantly boosts R&D, reduces cost and prototyping by enhancing the comprehension of customers' needs (Amro et al., 2025) (Rao, 2021) (Shrimal, 2024).

4.1.3 Demand forecasting Operations (topic_2)

The second most present topic, Demand forecasting Operations (15,4%), highlights the growing influence of predictive analytics and data driven decision making in operational planning and supply chain management with the help of AI. The literature evidence how the machine learning models outperform respect to traditional models dealing with complex, volatile data (Muth et al., 2025); global ML models enable accuracy and error control improvement, especially in heterogeneous and intermittent demands (Rožanec et al., 2021). AI driven models also support inventory management and supplier selection by analyzing risk factors that change quickly enabling cost reduction (Y. Wang et al., 2025) (Chehbi Gamoura et al., 2025). Looking at the after-sales context the AI and ML models use enhance behavioral forecast, like with XGBoost the maintenance visits prediction achieves higher retention and operative performance (Yıldırım et al., 2025). In general, the Tree based and hybrid ML enabling better inventory planning and reducing costs while maintaining quality (Fu et al., 2026). A limitation is present in context like EV charging demand from the availability, standardization and scalability difficulties (Rashid et al., 2024).

4.1.4 Data-Driven Mobility Governance (topic_3)

Another important topic identified by the LDA model is Data-Driven Mobility Governance (12,8%), which captures the relationship between tech innovation, regulatory framework and sustainability objectives. This information demonstrates the growing importance of policy and governance mechanisms in reshaping the automotive ecosystem, particularly in the context of electrification and energy transition. The literature highlights that the use of AI in explorative way enhances the existing products and process rather than enabling a radical innovation (Plantec et al., 2023). Analytics and business intelligence, if combined with knowledge sharing, can improve brand innovation

and organizational capabilities (Xu & Xu, 2025). The global value chains are influenced by the regulatory constraints and from data governance that change the dependencies between automotive players across regions (Stefanov & Schwabe, 2025). Some difficult challenges remain, like fragmented data infrastructure, complex relationship between AI and mobility market development such as electric vehicles (Gu et al., 2024).

4.1.5 Sustainable Product Design (topic_4)

Less frequent but still relevant are the topic of Sustainable Product Design (7,7%), where the literature highlight that the use of AI, Machine Learning within fuzzy logic and multi criteria models enable a better understanding of the consumers taking to a more aimed product design with specific solutions also in the EV section (Micus et al., 2025) (Mohammadian & Valilai, 2024) (Liu et al., 2024). The data driven approach combined with the analysis of the preferences also take to a batter responsiveness of the supply chain to the key attributes requirement of the consumer (Sathyan et al., 2021) (Chomachaei, 2025), the same increment of knowledge enhance the design of AI service directly on board reducing the perceived risk and increasing the satisfaction (Mathai, 2024).

4.1.6 Safety System (topic_5)

Inspecting the topic_5 called Safety System, that is the 12% of the distribution, emerge that deep learning and reinforcement learning in the autonomous driving take to new risks and scalability difficulties along all the process (Mirzarazi et al., 2024) (Kiran et al., 2022). The literature also underlines how a full cover assurance approach is needed along the whole process because of the high standard of security needed that are not satisfied by this technology (Ballingall et al., 2023) (Borg et al., 2019).

In the same environment also the regular software engineering results insufficient for autonomous vehicles. So, in general, a specified model must be built for those practices with high risk (Moukahal et al., 2020). Another important aspect of this topic is that the high percentage of black box reasoning in the AI and ML models reduces the trust and reliability of the costumers (Omeiza et al., 2022), the same pattern is also fundable in the communication systems (Tan et al., 2024).

4.1.7 User Experience Design (topic_6)

Finally, the smallest topic, User Experience Design (2,6%), is representative of a niche but strategically important area focused on human -machine relationships and costumer-centered industry. Particularly, the use of augmented reality leads to a better and more personalized experience for the customer and enhances the perception of the service, although the purchase intensions is not always maintained with the time (Poushneh & Vasquez-Parraga, 2024).

The general use of AI and Machine Learning approach enhances all the costumer process with better marketing, interaction across the touchpoints taking to higher value life cycle of the product (Chai & Islaim, 2021).

4.1.8 Decision Support (topic_7)

The Decision Support topic represents around 12% of the distribution. This theme reflects the operational and strategical applications of AI, including optimization algorithms and predictive models. Data driven systems that combine fuzzy logic, DEMATEL and entropy weighting achieve evaluation of complex environment, like design, life cycle sustainability (Kurrahman et al., 2025). The functional insight revealed from those models enable a more informed resource allocation and transformation planning (Çıkmak, 2025) (Kiatcharoenpol & Sirisawat, 2026) Hybrid AI technique consistently improve prediction, robustness enhancing decision-making processes (Salais-Fierro et al., 2020) (Sieberg et al., 2022). In parallel, ML tools help real time decision making in predictive maintenance and failure detection (Redondo et al., 2020).

4.1.9 Digital transformation (topic_8)

The most discussed topic in the texts is labelled as Digital transformation strategy and account proximately for 19,2% of the total publications. This element evidences the central role of organizational and strategic transformation in the AI adoption in the supply chain of the automotive sector. The same results also suggest that research attention is increasingly shifting from a mere technological development toward a boarder business and operational transformation process. Specifically, the changes are dominated by collaborative and circular data driven ecosystem, where the figure evolves dynamically to achieve value co-creation (Kortus et al., 2025) and to high the transparency of SDG

related reporting (Lin, 2026). From an operative prospective the papers in this topic explain that the studied sector is experiencing a re-dimensioning of the AI supervising role's with an higher importance in the data-driven competence (Colombari et al., 2024), other texts highlights how the use of the modern technology is directly related to the capability of the industry to adapt and develop (Bodendorf & Franke, 2024) mostly in the PMI where the industry 4.0 makes an higher difference (Dwivedy et al., 2025) (Fodor et al., 2026).

Another argument is regarding the supply chain and the use of IoT and big data to enhance the resilience and reactivity (Amer et al., 2026) (Nayal et al., 2024), also the KPI and decisional process reliability can be higher with the same technologies (El Kihel et al., 2023). In CRM and big data analysis also, the AI play a crucial role, by elevating the costumer centered approach by improving marketing performance (Hu & Basiglio, 2024). From a strategic perspective, the literature highlights the need of a continues advancement, implementation of technology and competence (Lopez-Vega & Moodysson, 2023).

So, in general, the evolution of AI, technology and their implementation is fundamental to high the reliability, efficiency and the costumer experience (Gupta et al., 2021).

All the results evidence a diversified research landscape where AI is used as a technological tool but also as a catalyst for organizational transformation, operation optimization and strategic decision-making within the automotive sector.

5 Research agenda

The following table reports the possible future analysis to do by looking at the present gap in the literature under the 3 main themes revealed in the actual knowledge.

Theme	Description	Possible Research Question
Business application (Industrial Innovation Management, Demand Forecasting Operations, Data-Driven Mobility Governance, Decision Support, Digital Transformation Strategy)	AI implementation in product development, innovation process, customer analysis and forecast	RQ1: Which are the main organizational capabilities required to an automotive industry to apply the transition to AI data-centered company?
		RQ2: How can the data-driven ecosystem create new revenue sources in the automotive industry?
		RQ3: How do regulatory constraints limit the development of AI data-driven value prepositions in the mobility sector?
		RQ4: Which are the organizational barriers that limit the adoption of AI decision support system in automotive companies?
		RQ5: How AI enables the reshape of relationship between decision systems and external stakeholder?
Sustainability (Battery Fault Diagnostics, Data-Driven Mobility Governance, Sustainable Product Design)	AI driven sustainable design and enhance product performance and efficiency	RQ1: How the customer's behavior and the digital engagement change the adoption of sustainable automotive solutions?
		RQ2: How a new player in a complex sector like automotive can arrive to implement a data driven centered approach and achieve the optimization in the energy consumption?
		RQ3: How can AI support the integration of sustainability criteria into early-stage product design process in the automotive industry?
		RQ4: How can predictive maintenance systems reduce waste in electric vehicle operations and energy consumption?
		RQ5: Which are the main challenges in implementing ESG objectives in a AI data driven automotive environment?

Theme	Description	Possible Research Question
User Experience (User Experience Design, Safety Systems)	AI can high the personalization, human-machine interaction, the safety and reliability	RQ1: How the costumer perceived the implementation of AI in their vehicle?
		RQ2: How are the advantages given by AI effectively perceived by the consumer? The perception changes in automotive sector?
		RQ3: How can the companies improve the user trust and regulatory acceptance in autonomous driving?
		RQ4: How the iper-personalization influences the trust of the consumers in intelligent vehicle interfaces?
		RQ5: Which are the main factors that influence the costumers trust toward safety perception in autonomous driving environment?

The following mind map summary the present study findings to evaluate conjunctively the proposed research agenda and the actual knowledge:

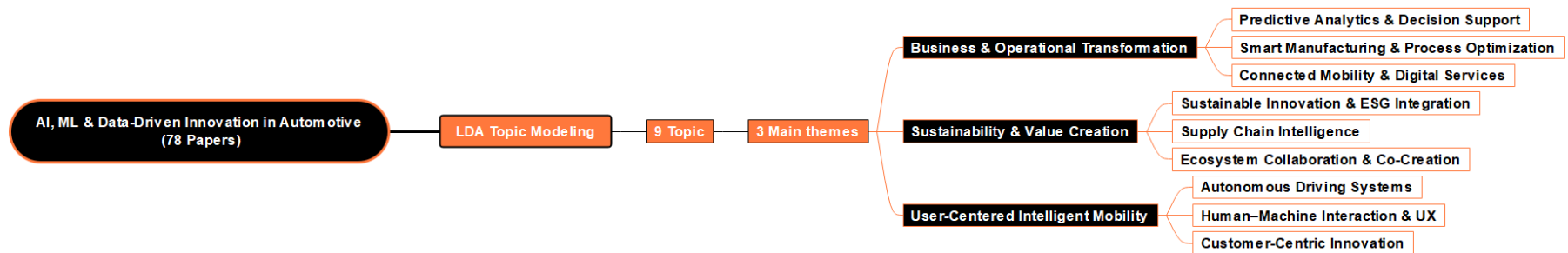


Figure 13 Mind map

6 Discussion

The precedent overview given by the result of the topic modelling analysis gives the possibility to achieve valuable insight into the rapid evolution of the AI implementation in automotive industry and highlights 3 main macro areas. First, Business Model Influences comprehends topics related to “Industrial Innovation Management”, “Demand Forecasting Operations”, “Data-Driven Mobility Governance”, “Decision Support”, “Digital Transformation Strategy” evidencing how AI is reshaping the value preposition. Second, Sustainability contains topics like “Battery Fault Diagnostics”, “Data-Driven Mobility Governance”, “Sustainable Product Design” highlighting the importance of AI in enhancing energy efficiency, environmental responsibility and circularity. Third, User experience includes “User Experience Design” and “Safety Systems” emphasizing the fundamental role of human-machine interaction, trust and safety in the implementation of AI automotive technology.

6.1 Business Model Influences

The business model influences are the main area highlighted from the analysis of the actual literature knowledge. Prior studies evidence different gaps, first (Vacarescu et al., 2025) highlight the absence of integrated frameworks linking Business model transformation and AI. Marketing oriented studies (Ghantous & Alnawas, 2021) highlight the insufficient exploration in AI driven ecosystem and platform-based value creation in industrial markets. These gaps has been addressed in this research by analysing and taking into exam different research that, together, give a more complete understanding of the topic reducing the fragmentation of the present literature; specifically, the paper under the topics “Industrial Innovation Management” and “Digital Transformation Strategy” express how AI can support the business model evolution in the automotive sector. Similarly, research on forecast and analytics (Ranabhatt et al., 2025) evidence the limitations in combining predictive models with the effective management and value chain of a company. The papers under the topics “Demand Forecasting Operations”, “Decision Support” fill out this gap concluding that AI and hybrid algorithm can effectively enhance accuracy and enabling more informed managerial decision improving all the value chain.

6.2 Sustainability

Another important research gap concerns the use of artificial intelligence in relation to sustainability and environmental objectives. Recent studies (Li et al., 2021) emphasizes the low integration of AI-driven technology and circular economy and the lack of holistic frameworks. Similarly, other research (Carvalho et al., 2024) express the insufficient knowledge present about how technologies can support environmental sustainability and mobility system efficiency.

This literature review directly addresses these gaps. In “Sustainable Product Design” topic is demonstrated how AI frameworks (as SAPAD-FQFD with PSO-RF models) can align the user needs with the sustainability criteria and enhance the general performance.

Moreover, “Battery Fault Diagnostics” topic highlights how the advanced data driven models can effectively monitor and optimize battery life. Those techniques combined with classical algorithms like clustering and statistical models can directly support the sustainability goals and high the efficiency incrementing the life cycle management.

Finally, “Data driven Mobility Governance” express how data analytics can improve the energy efficiency and operational sustainability in mobility system, providing actionable insights for greener mobility planning.

6.3 User Experience

Furthermore, in the User Experience ambient recent studies evidence that there is gaps related to human-AI interaction and trust of consumer in those systems. Regress research (Cescon & Peruzzini, 2026) examine the lack of user-centric design that balances automation, perceived control and usability. Another paper (Ebel et al., 2021) highlights a limited knowledge of trust and behavioral response drivers in the automotive industry.

The present study addresses this gaps individuating two main topics: “User experience Design”, where is evidenced that hybrid technology can high the satisfaction, perceived quality and usability by design product that make the costumer feels the center of the decisional process, and topic as “Safety system”, where papers says that trust and acceptance of systems such AI or hybrid model in an environment that present high risk consequences are harder and require a specific system development,

7 Theoretical and Managerial Contribution

This study provides several contributions to both academic research and managerial practice in the artificial intelligence environment and innovation within the automotive industry.

7.1 Managerial contributions

The findings evidence the strategic importance of digital transformation as a main element of innovation and driver of performance in the automotive industry. The results of LDA highlight that artificial intelligence, data-driven approaches and ML are increasingly used to enhance decision making process, optimize investments in data infrastructure, digital capabilities and predictive analytics. Those characteristics enable continues innovation and operational resilience.

The study contributes to strategic decision-making and policy formulation by explaining the emerging research areas related to sustainability, regulatory framework and electrification. These insights can help industry stakeholders and policymakers in creating effective strategies for the implementation of technology in a sustainable path to achieve long-term competitiveness.

Practical advice of application follows;

First, the study highlights the necessity of shifting from isolated AI applications to an integrated, cross-functional, strategies planning. Firms that adopt this type of approach can achieve a predictive and prescriptive business model instead of reacting and descriptive attitude.

Second, the research underlines the fundamental importance of aligning AI investments with sustainability objectives to encourage a circular economy from the design to the end-of-life cycle.

Third, the present conclusion emphasizes the central role of user-centric design and safety assurance to obtain market acceptance in case of AI-enabled vehicles. The principal suggestion is to invest not only in performance improvements but also in trust and transparency between human and machine interaction.

Using practical terms, the following of the advices from this study makes a company in automotive industry able to evolve toward and achieve better decision-making, forecast, product design and development. The evolution of this sector and the application of AI in it represents a shift that must be followed to maintain or to achieve competitive advantage in the market.

7.2 Theoretical contributions

The research offers a structured and data-driven view of the main thematic emerging in the literature. By using topic modeling techniques on a reviewed database of publications, the study focalizes nine different domains that together describe the actual state of the knowledge in the field. The systematic classification also contributes to reducing the general conceptual fragmentation and giving a clear framework for evaluating the diverse applications of artificial intelligence across operational, strategic and technological dimensions.

More importantly, the present research extends the existing literature on how AI enables business models in automotive. In line with prior studies highlight the strategic role that AI has in industrial changing (Ghantous & Alnawas, 2021); this thesis clarifies how AI is fundamental as technological tool but also as enabler of value creation, organizational design and ecosystem development.

The output of analysis contributes by putting in relationship gaps identified in the recent literature. Specifically, the forecast and decision-making context (Ranabhatt et al., 2025; Vacarescu et al., 2025) has expanded by showing how data-driven and AI models can outperform on this argument. Similarly, sustainability research emphasizes a link between digital technologies and environmental performance (Carvalho et al., 2024; Li et al., 2021) the present results integrate and contribute to the topic.

Lastly, prior papers (Cescon & Peruzzini, 2026; Ebel et al., 2021) on intelligent mobility evidence the need of improvement in human-centric design products and trust, necessity which through the present paper is addressed by combining insights of “User experience design” and “Safety System” topics.

7.3 Replicable workflow

A last and fundamental contribution is the knime workflow available and exportable on knime hub (under the profile of alessandrocaros in the Thesis folder and in the Thesis 3 file). This workflow can be used as basis of every systematic literature review with topic modelling algorithms; every node is explained in the comments and can be understood even if the user is new on knime. This instrument gives to any individual the opportunity to perform a rapid and general overview of the literature framework on every theme, in a bunch of minutes with non-excellent accuracy, but enough to achieve an overall idea of the discussed topic.

7.3.1 Workflow characteristics

Just by selecting the path in the excel/csv reader node different types of paper's provider files (like Scopus, IEEE Explorer, Google scholar or others) are importable; the flow is automated for those providers, otherwise just by renaming the column with the column renamer node is possible to apply the automation. Along the workflow are present different check boxes to ensure the right computing of the program (the instructions to check are always in the comments of each node). To use text mining processing the text mining libraries must be installed in knime. In the last part of the workflow in the section "Python LDA Visualization" the exportation of the file is performed to achieve a Json complete file with all the info, that is done externally on Visual Studio Code with the script contained in the Appendix paragraph (11.2). The use of external resources can be omitted by using the block under the visualization where a python script already is settled and runnable (in appendix (11.5)). To obtain the visualization without exiting from Knime replace the output of the python script to the node called "Python Script (1=>1) (deprecated)" (the one that create the HTML visualization (in the appendix (11.4)). If a permanent link for the online visualization is needed just use the node that export the HTML and publish it in a repository on GitHub or similar that will provide automatically a link that can be past as URL.

Also the bibliography option is implemented: the last node, group by, in the Bibliography box give as output a long string where all the bibliography in APA format is in one cell

and ready to be copy and pasted in the study. If a table where for each paper the source in APA format is needed just open the meta node and click on the last component.

8 Limitations

The present study is subject to several limitations given by data selection, methodologic decision and model interpretation. Rigorous procedures were adopted through all the work to ensure reliability and transparency; given this the results should be interpreted considering the constraints inherent in bibliometric and text mining approaches.

The dataset may not fully capture the entire body of the knowledge of artificial intelligence in driving business model implementation in automotive sector. The selecting process ultimately resulted in a final sample of 78 papers, which, although suitable for exploratory and analytics purposes, represents a relatively limited subset of the boarder research landscape. The preset search is based on publication from 2020 to the present data (05/2026), meaning that, in a world that has an extremely high evolutionary speed, perhaps, would be outdated in a few years.

Finally, the research is based on academic publications written in English, characteristic that may introduce language bias and limit the representation of research in non-English speaking regions. This constrains is more relevant where in region with a different linguistic tradition a technological development occurs (like China).

The adoption of LDA in the research enables systematic and scalable analysis but also involves several subjective and technical decisions that influence the results. For example, the preprocessing of the text requires the application of tokenization, stop word filer and normalization that can reduce all the little faces of a meaning to a unique strict sense.

The generated topics do not represent definitive thematic categories but rather statistical pattern of word co-occurrence in the dataset. The labelling of topics is influenced by researcher judgement, characteristics that introduce a degree of subjectivity.

9 Conclusion

The research evaluates the role of artificial intelligence, data-drive approaches and machine learning in driving innovation within the automotive industry by using topic modeling techniques on a systematically selected database composed by 78 scientific publications. The output of the analysis reveals a dynamic and evolving landscape characterized by a convergence in technological innovation, organizational transformation and strategic decision making.

The analysis highlights that the most discussed topics in the actual literature are the digital transformation and operational optimization. Emerging areas like sustainable product design, battery diagnostics and user experience design indicate the growing complexity of the automotive sector and the need for interdisciplinary approaches to innovate in the ecosystem.

The integration of AI in long-term business models, sustainability strategies and user centered design process remains several challenges and requires further investigations. The address of the nominated will be essential to fully realize the potential of AI and new analytical technologies in the automotive industry.

Overall, the present study provides a comprehensive and structured overview of the actual knowledge in the field and place the bases for the next studies that aims to advance the framework of technological innovation, operational efficiency and strategic competitiveness in the automotive sector.

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11 Appendix

11.1 Python script to query semantic scholar through API

```
import knime.scripting.io as knio
import pandas as pd
import requests
import time

input_df = knio.input_tables[0].to_pandas()
results = []
s2_fields =
"title,authors,year,externalIds,abstract,publicationTypes,citationCount"

for index, row in input_df.iterrows():
    title = str(row.get('Title', '')).strip()
    authors_raw = str(row.get('Authors', '')).strip()
    year_raw = row.get('Year', None)

    if not title or title.lower() == 'nan':
        continue

    try:
        if pd.notna(year_raw):
            year = str(int(float(year_raw)))
        else:
            year = ""
    except (ValueError, TypeError):
        year = ""

    first_author = authors_raw.split(',')[0].strip() if authors_raw else ""

    query = f"{title} {first_author}".strip()

    url = "https://api.semanticscholar.org/graph/v1/paper/search"
    params = {
        "query": query[:300],
        "fields": s2_fields,
        "limit": 1
    }

    if year:
        params["year"] = year

    row_result = {
```

```

        "Authors": authors_raw,
        "Title": title,
        "Year": year,
        "DOI": "",
        "Abstract": "",
        "Author Keywords": "",
        "Index Keywords": "",
        "Document Type": "",
        "Article Citation Count": 0
    }

    max_tentativi = 3
    for tentativo in range(max_tentativi):
        try:
            response = requests.get(url, params=params)

            if response.status_code == 200:
                data = response.json()
                if 'data' in data and len(data['data']) > 0:
                    paper = data['data'][0]

                    ext_ids = paper.get('externalIds', {})
                    doi = ext_ids.get('DOI', '') if isinstance(ext_ids,
dict) else ''

                    pub_types = paper.get('publicationTypes', [])
                    doc_type = ", ".join([pt for pt in pub_types if pt])
                    if pub_types else ''

                    authors_list = paper.get('authors', [])
                    authors_formatted = ", ".join([a.get('name', '') for
a in authors_list]) if authors_list else authors_raw

                    row_result.update({
                        "Authors": authors_formatted,
                        "Title": paper.get('title', title),
                        "Year": str(paper.get('year', year)),
                        "DOI": doi,
                        "Abstract": paper.get('abstract', ''),
                        "Document Type": doc_type,
                        "Article Citation Count":
int(paper.get('citationCount', 0) or 0)
                    })
                    break

            elif response.status_code == 429:
                pausa = 5 * (tentativo + 1)

```

```

        print(f"Riga {index}: Rate limit. Attendo {pausa} secondi
(Tentativo {tentativo + 1}/{max_tentativi})...")
        time.sleep(pausa)
    else:
        print(f"Riga {index}: Errore API {response.status_code}")
        break

except Exception as e:
    print(f"Errore di rete riga {index}: {e}")
    break

results.append(row_result)

time.sleep(3)

output_df = pd.DataFrame(results)
colonne_testo = ["Authors", "Title", "Year", "DOI", "Abstract", "Author
Keywords", "Index Keywords", "Document Type"]
for col in colonne_testo:
    output_df[col] = output_df[col].fillna("").astype(str)

output_df["Article Citation Count"] = pd.to_numeric(output_df["Article
Citation Count"], errors='coerce').fillna(0).astype('int64')

knio.output_tables[0] = knio.Table.from_pandas(output_df)

```

11.2 Visual studio code script to merge the data for the PyLDAviz

```
import pandas as pd

import json

import numpy as np

import os


phi = pd.read_csv('phi.csv')

theta = pd.read_csv('theta.csv')

vocab_freq = pd.read_csv('vocab_freq.csv')

doc_len = pd.read_csv('doc_len.csv')


print("File upload done")


lista_vocab = vocab_freq['vocab'].astype(str).tolist()


phi_matrix = phi.reindex(columns=lista_vocab).fillna(0)


theta_matrix = theta.select_dtypes(include=[np.number]).values.tolist()


phi_final = phi_matrix.values.tolist()


doc_lengths = doc_len['doc.length'].tolist()
```

```
term_frequencies = vocab_freq['term.freq'].tolist()
```

```
lda_json_data = {  
    "phi": phi_final,  
    "theta": theta_matrix,  
    "doc_len": doc_lengths,  
    "vocab": lista_vocab,  
    "term_freq": term_frequencies  
}
```

```
file_output = 'data_progetto.json'
```

```
with open(file_output, 'w', encoding='utf-8') as f:
```

```
    json.dump(lda_json_data, f, ensure_ascii=False, indent=2)
```

```
print(f"\nDONE")
```

```
print(f"Il file '{file_output}' has been created in the folder:")
```

```
print(os.getcwd())
```

```
print(f"\nVerifying data coherence:")
```

```
print(f"- Topic's Number (in phi): {len(phi_final)}")
```

```
print(f"- Word's Number (in vocab): {len(lista_vocab)}")
```

```
print(f"- Document's number (in theta): {len(theta_matrix)}")
```

```
print(f"- Document's number (in doc_len): {len(doc_lengths)}")
```

```

if len(theta_matrix) != len(doc_lengths):

    print("ATTENTION: The number of documents in theta and in doc_lenght is not the
same!")

if len(lista_vocab) != len(phi_final[0]):

    print("ATTENTION: The number of documents in vocab and in phi is not the same!")

```

11.3 Python script to create the LDA visualization

```

import numpy as np

import pandas as pd

import json

import pyLDAvis

data_input = json.loads(input_table['json'][0])

phi = np.array(data_input['phi'])

theta = np.array(data_input['theta'])

phi = phi / phi.sum(axis=1).reshape(-1, 1)

theta = theta / theta.sum(axis=1).reshape(-1, 1)

movies_model_data = {

    'topic_term_dists': phi,

    'doc_topic_dists': theta,

    'doc_lengths': data_input['doc_len'],

```

```

'vocab': data_input['vocab'],

'term_frequency': data_input['term_freq']
}

print('Topic-Term shape: %s' % str(phi.shape))

print('Doc-Topic shape: %s' % str(theta.shape))

movies_vis_data = pyLDAvis.prepare(**movies_model_data, sort_topics=False)

output_table = pd.DataFrame({"pyldavis_html":
[pyLDAvis.prepared_data_to_html(movies_vis_data)]})

```

11.4 Python script to export the visualization in html

```
import pandas as pd

import os

html_content = input_table['pyldavis_html'][0]

cartella_tesi = r'C:\Users\Alessandro Carosi\luiss\Magistrale\Thesis'

nome_file = "Visualizzazione_LDA_Interattiva.html"

percorso_completo = os.path.join(cartella_tesi, nome_file)

try:

    if not os.path.exists(cartella_tesi):

        os.makedirs(cartella_tesi)

        print("Cartella creata perche' non esistente.")

    with open(percorso_completo, "w", encoding="utf-8") as file:

        file.write(html_content)

    print("SUCCESSO: Il file e' stato salvato correttamente.")

    print("Percorso file: " + percorso_completo)

except Exception as e:

    print("ERRORE DURANTE IL SALVATAGGIO:")
```

```
print(str(e))
```

```
output_table = input_table.copy()
```

11.5 Python script to remain in knime

```
import pandas as pd

import os

html_content = input_table['pyldavis_html'][0]

cartella_tesi = r'C:\Users\Alessandro Carosi\luiss\Magistrale\Thesis'

nome_file = "Visualizzazione_LDA_Interattiva.html"

percorso_completo = os.path.join(cartella_tesi, nome_file)

try:

    if not os.path.exists(cartella_tesi):

        os.makedirs(cartella_tesi)

        print("Cartella creata perche' non esistente.")

    with open(percorso_completo, "w", encoding="utf-8") as file:

        file.write(html_content)

    print("SUCCESSO: Il file e' stato salvato correttamente.")

    print("Percorso file: " + percorso_completo)

except Exception as e:

    print("ERRORE DURANTE IL SALVATAGGIO:")
```

```
print(str(e))
```

```
output_table = input_table.copy()
```

11.6 Summary table

Paper	Method	Context	Focus / Topic	Key Findings
Kurrahman, T., Tsai, F. M., Sethanan, K., Lim, M. K., & Tseng, M.-L. (2025). Data-driven life cycle assessment of the automobile industry in Indonesia: Identifying circular supply chain enablers. Resources, Conservation and Recycling. (https://doi.org/10.1016/j.resconrec.2025.108338)	Mixed-method data-driven approach (EWM, FDM, EFA, FDEMATEL)	Indonesian automobile industry	Circular supply chain (CSC) enablers	Prioritizing eco-design, product refurbishment, and waste management systems is essential for enhancing lifecycle circularity and sustainability.
Yang, M., Zhu, H., & Guo, K. (2020). Research on manufacturing service combination optimization based on neural network and multi-attribute decision making. Neural Computing and Applications. (https://doi.org/10.15837/ijccc.2025.4.7123g/10.1007/s00521-019-04241-6)	Machine learning model using multi-attribute decision-making and BP neural network	Automotive manufacturing and service operations	Product design evaluation and service innovation decision support	Neural network-based decision model achieved high accuracy and supports optimized product design and service planning decisions
Plantec, Q., Deval, M.-A., Hooge, S., & Weil, B. (2023). Big data as an exploration trigger or problem-solving patch: Design and integration of AI-embedded systems in the automotive industry. Technovation. (https://doi.org/10.1016/j.technovation.2023.102763)	Mixed-method approach using patent analysis and qualitative interviews	Automotive new product development and AI system integration	Data-driven innovation and AI adoption strategies in vehicle development	Firms mainly use AI to enhance existing systems incrementally, with organizational and data challenges limiting radical innovation

Paper	Method	Context	Focus / Topic	Key Findings
<u>Kortus, L., Gutmann, T., & Kreutzer, M. (2025). Unearthing role dynamics within innovation ecosystem emergence: A case study of data-driven circularity. R&D Management. (https://doi.org/10.1111/radm.12741)</u>	Ethnographic qualitative case study using observations, interviews, and document analysis	Innovation ecosystem for circular economy solutions in the German automotive industry	Role evolution and collaboration dynamics in data-driven circular economy ecosystems	Partner roles remain task-stable but behaviorally dynamic; role reconfiguration driven by internal/external dynamics enhances collaborative value creation for circular economy solutions
<u>Lin, W. L. (2026). The influence of circular economy on firm orientation toward sustainable development goals: The moderating role of technological innovation in the automotive industry. Sustainable Development. (https://doi.org/10.1002/sd.70278)</u>	Quantitative longitudinal study using dynamic panel data analysis (System GMM)	Global automotive firms (2016–2022)	Impact of circular economy practices and technological innovation on SDG disclosure performance	Circular economy practices significantly improve the depth and breadth of SDG disclosures, while technological innovation strengthens these effects and enhances sustainability reporting transparency

Paper	Method	Context	Focus / Topic	Key Findings
Xu, Z., & Xu, X.-Y. (2025). Unlocking the secrets of business intelligence and analytics in driving brand innovation: Insights from an empirical study using SEM and fsQCA. Cogent Business & Management. (https://doi.org/10.1080/23311975.2025.2494066)	Survey-based quantitative study using PLS-SEM and fsQCA	Traditional automobile manufacturing firms in China	Business intelligence and analytics for brand innovation	Combined BI&A utilization, knowledge sharing, and absorptive capacity jointly drive higher brand innovation performance
Nathan, B. S., Siva Reddy, B. V., Sastry, C. C., Krishnaiah, J., & Eswaramoorthy, K. V. (2024). Innovative framework for effective service parts management in the automotive industry. Frontiers in Mechanical Engineering. (https://doi.org/10.3389/fmech.2024.1361688)	Mixed-method approach (data analysis + predictive modeling + rule-based optimization)	Automotive service parts management (Thailand automotive industry)	Inventory optimization and demand forecasting for service parts	Demand classification and decision rules reduce costs while improving service parts availability and operational efficiency
Slouma, S., Romdhane, M. B.-S., & Skander-Mustapha, S. (2026). Current state, challenges, recommendations and prospects of machine learning application in fuel cell based hybrid electric cars: A comprehensive overview. Journal of Sustainable Development of Energy, Water and Environment Systems. (https://doi.org/10.13044/j.sdewes.d13.0641)	Literature review and comparative analysis of machine learning applications	Hydrogen fuel cell electric vehicles and energy management systems	Energy management optimization, predictive maintenance, fault diagnosis, traffic integration	ML-based energy systems reduce costs up to 28%, improve fuel economy 5.7%, and cut emissions and degradation; gaps in predictive maintenance and safety remain

Paper	Method	Context	Focus / Topic	Key Findings
<u>Chirinda, G. P., Matope, S., & Nagel, M. (2024). Development of a material selection decision support system for an automotive application. Discover Mechanical Engineering.</u> https://doi.org/10.1007/s44245-024-00071-6	Experimental study + decision support system development (web-based multi-criteria model)	Automotive manufacturing (rooftop tent production using aluminium alloys)	Material selection optimization for lightweight automotive components	AA1050 selected as optimal alloy balancing cost, formability, and sufficient mechanical performance
<u>Kamble, S. S., Gunasekaran, A., Ghadge, A., & Raut, R. (2020). A performance measurement system for industry 4.0 enabled smart manufacturing system in SMMEs: A review and empirical investigation. International Journal of Production Economics.</u> https://doi.org/10.1016/j.ijpe.2020.107853	Empirical study + Exploratory research (EFA + CFA)	Automotive SMMEs (auto-component manufacturing, India; Industry 4.0 smart manufacturing systems)	Performance measurement of smart manufacturing investments	Identified and validated 10 performance dimensions (cost, quality, flexibility, time, integration, productivity, diagnosis, computing, social & ecological); SMS improves competitiveness over traditional systems

Paper	Method	Context	Focus / Topic	Key Findings
<u>Colombari, R., Neirotti, P., & Berbegal-Mirabent, J. (2024). Disentangling the socio-technical impacts of digitalization: What changes for shop-floor decision-makers? International Journal of Production Economics. (https://doi.org/10.1016/j.ijpe.2024.109377)</u>	Exploratory qualitative study (semi-structured interviews, grounded theory, thematic analysis)	Automotive manufacturing shopfloors (Italy & Spain Tier-1 suppliers and OEMs)	Digital transformation; shop-floor decision-making; socio-technical change	Digitalization leads to polarized decision-making: decentralization to team leaders for operations and centralization to production managers for continuous improvement; supervisors' role diminishes.
<u>Micus, C., Homola, D., Prommegger, B., Thaetz, J., & Krcmar, H. (2025). Leveraging BMW fleet data for customer integration in product development. Results in Engineering. (https://doi.org/10.1016/j.rineng.2025.105498)</u>	Design science research utilizing big data analytics, vehicle GPS telemetry, and Gaussian Mixture Model (GMM) clustering.	Global automotive manufacturing and parking assistance systems within the European Union, United States, and Chinese markets.	Utilizing actual vehicle usage data to classify parking scenarios and enhance customer-oriented product development and innovation.	Analyzing real-world telemetry identifies market-specific parking behaviors, enabling manufacturers to prioritize high-value functions and optimize development resources.

Paper	Method	Context	Focus / Topic	Key Findings
Liu, Q., Jia, M., & Xia, D. (2023). Dynamic evaluation of new energy vehicle policy based on text mining of PMC knowledge framework. Journal of Cleaner Production. (https://doi.org/10.1016/j.jclepro.2023.136237)	Big data-driven approach using policy informatics, PMC index modeling, text mining, and ontology semantic methods for dynamic evaluation.	China's national New Energy Vehicle (NEV) industry policies across five developmental phases from 2001 to 2021.	Assessing the dynamic evolution, structural consistency, and effectiveness of NEV policies to identify industrial development gaps.	Declining policy consistency and structural imbalances require enhanced inter-governmental cooperation and supply-side tools to support high-quality growth.
Dwivedy, M., Pandit, D., & Khatter, K. (2025). Assessment of smart manufacturing readiness for small and medium enterprises in the Indian automotive sector. Sustainability. (https://doi.org/10.3390/su17188096)	Adapted IMPULS framework utilizing a 43-indicator questionnaire, hierarchical clustering, and SWOT analysis for SME-specific empirical validation.	Indian automotive sector, targeting resource-limited Tier-2 and Tier-3 Small and Medium Enterprises (SMEs).	Contextualizing smart manufacturing readiness models for smaller firms facing budget, expertise, and infrastructure constraints.	The adapted framework enables resource-constrained SMEs to identify diagnostic gaps across six dimensions and implement actionable pathways for smart manufacturing adoption.

Paper	Method	Context	Focus / Topic	Key Findings
<u>Heisig, P., Ogaza, M. A., & Hamraz, B. (2020). Information and knowledge assessment: Results from a multinational automotive company. International Journal of Information Management.</u> https://doi.org/10.1016/j.ijinfomgt.2020.102137	Information and Knowledge Assessment (IKA), utilizing open-ended questionnaires, manual qualitative coding (classification), and expert mapping.	Engineering division of a multinational automotive supplier across six European sites (Austria, Czech Republic, France, Germany, Italy, Turkey).	Identifying information and knowledge needs of engineers and evaluating how well existing internal resources cover those needs.	Granular content categories provide better insights for information system design than broad system-level analysis; the IKA method is resource-efficient and successfully bridges the gap between theoretical auditing and industrial practice.
<u>Amer, Y., Soufali, A., & Zaghwan, A. (2026). A digital twin-based framework for predictive quality assurance and supply chain resilience in the automotive industry. Advanced Engineering Informatics.</u> https://doi.org/10.1016/j.aei.2025.103969	Conceptual framework (7D model) validated via case study and digital twin simulation.	Automotive supply chain and manufacturing operations.	Predictive quality assurance (PQA) and supply chain resilience (SCR).	The IATF-aligned framework enhances resilience and traceability by enabling proactive quality interventions and real-time disruption monitoring.

Paper	Method	Context	Focus / Topic	Key Findings
<u>Singh, S., Yadav, B., & Batheri, R. (2023). Industry 4.0: Meeting the challenges of demand sensing in the automotive industry. IEEE Engineering Management Review. (https://doi.org/10.1109/EMR.2023.3292331)</u>	AI and machine learning model incorporating demand signals, social listening, and offline consumer traffic data.	Automotive industry demand management and vehicle sales.	Demand sensing and short-term and long-term forecasting.	Industry 4.0 technologies and multi-channel signals improve demand prediction by addressing traditional forecasting shortcomings.
<u>Gu, J., Wu, Z., Song, Y., & Nicolescu, A.-C. (2024). A win-win relationship? New evidence on artificial intelligence and new energy vehicles. Energy Economics. (https://doi.org/10.1016/j.eneco.2024.107613)</u>	Econometric study using wavelet-based quantile-on-quantile regression (QQR) with wavelet decomposition	Electric vehicle market and artificial intelligence in automotive industry	AI influence on electric vehicle adoption and dynamic bidirectional interactions	AI–EV relationship is nonlinear and time-varying; short-term mixed effects and crowding-out exist; long-term interactions weaken, no stable win-win relationship
<u>Stefanov, A., & Schwabe, J. (2025). Global value chains for digital automotive products: The integration of navigation systems into Chinese cars in Europe. Geographica Helvetica. (https://doi.org/10.5194/gh-80-289-2025)</u>	Mixed-method approach (case studies + 43 interviews; GVC/GPN analysis)	Automotive industry (Chinese EV OEMs in European navigation systems market)	Digital value chains, navigation systems, regulatory impact on value creation	EU regulations and data constraints force Chinese OEMs to rely on European/US providers, reshaping value chain governance and dependencies

Paper	Method	Context	Focus / Topic	Key Findings
Rashid, M., Elfouly, T., & Chen, N. (2024). A comprehensive survey of electric vehicle charging demand forecasting techniques. IEEE Open Journal of Vehicular Technology. (https://doi.org/10.1109/OJVT.2024.3457499)	Systematic literature survey (taxonomic + comparative analysis of ML and probabilistic models)	Electric vehicle charging infrastructure	EV charging demand forecasting	ML methods perform well with real data but face data scarcity, lack of standards, and scalability issues
Wagino, Capilayan, M. A., Setiyo, M., Tymofiiiv, V., Putra, D. S., Yuvenda, D., & Koto, R. D. (2025). Smart and sustainable mobility: A systematic review of Industry 4.0 applications in automotive sustainability, alternative fuels, and EV charging systems. Automotive Experiences. (https://doi.org/10.31603/ae.13431)	Systematic review (PRISMA + bibliometric + content + scientometric analysis)	Automotive sustainability, alternative fuels, EV charging infrastructure	Industry 4.0 integration in sustainable mobility systems	AI, IoT, blockchain enable smart infrastructure, but standardization and interdisciplinary gaps remain
Akhtar, P., Ghouri, A. M., Ashraf, A., Lim, J. J., Khan, N. R., & Shuang, M. A. (2024). Smart product platforming powered by AI and generative AI: Personalization for the circular economy. International Journal of Production Economics. (https://doi.org/10.1016/j.ijpe.2024.109283)	Quantitative cross-sectional survey using SEM	Chinese automotive manufacturing industry	AI-driven smart product platforming for circular economy	AI/GAI-enabled SPP, big data analytics, and machine learning jointly enhance personalized eco-friendly product design and circular economy performance, with SPP acting as key mediator.

Paper	Method	Context	Focus / Topic	Key Findings
Karaman, E., Cilan, C. A., & Bozdogan, H. (2023). Hybridized model selection with Gifi system for categorical data using the genetic algorithm and information complexity. Electronic Commerce Research and Applications. (https://doi.org/10.1016/j.elerap.2022.101221)	Hybrid modeling approach (CATREG + genetic algorithm + ICOMP + neural network classification)	Cross-disciplinary datasets including e-commerce automotive market and behavioral health	Categorical data modeling and feature selection using AI optimization techniques	GA-enhanced CATREG with ICOMP improves variable selection, avoids local minima, and yields more accurate and interpretable classification performance.
Mirzarazi, F., Danishvar, S., & Mousavi, A. (2024). The safety risks of AI-driven solutions in autonomous road vehicles. World Electric Vehicle Journal. (https://doi.org/10.3390/wevj15100438)	Comparative analysis, theoretical review, and standards mapping of DNN safety risks	Autonomous driving (ADAS, autonomous road vehicles)	Safety risks of deep neural networks in autonomous driving systems	DNN classifiers introduce significant safety risks across design, training, and deployment; alignment with automotive safety standards is insufficient without additional safeguards
Moukahal, L. J., Elsayed, M. A., & Zulkernine, M. (2020). Vehicle software engineering (VSE): Research and practice. IEEE Internet of Things Journal. (https://doi.org/10.1109/JIOT.2020.3001026)	In-depth literature review and comparative analysis of software engineering processes	Connected Autonomous Vehicles (CAVs), automotive IoT software systems	Vehicle software engineering processes, tools, standards, and development practices	Traditional software engineering models are inadequate for CAVs; dedicated VSE models and best practices are required

Paper	Method	Context	Focus / Topic	Key Findings
<u>Sterk, F., Stocker, A., Heinz, D., & Weinhardt, C. (2024). Unlocking the value from car data: A taxonomy and archetypes of connected car business models. Electronic Markets. (https://doi.org/10.1007/s12525-024-00692-5)</u>	Sequential mixed-method taxonomy development, structured literature review, cluster analysis, expert interviews	Connected cars, automotive data ecosystems	Data-driven connected car business models and archetypes	Identifies taxonomy and seven archetypes showing how vehicle data enables diversified connected car business models
<u>Wang, T., Liu, Z., Chen, W., Luo, T., Sun, M., Liu, X., & Liu, Y. (2025). A multi-dimensional driving condition recognition method for data-driven vehicle noise and vibration platform based on speed separation strategy and deep learning model. Mechanical Systems and Signal Processing. (https://doi.org/10.1016/j.ymssp.2025.113001)</u>	Mixed-method approach (data parsing + regression + deep learning model: IResNet)	Automotive NVH testing and vehicle performance evaluation (electric vehicles)	Digital transformation of NVH condition recognition	Integrated speed-separation and IResNet approach achieves 90% accuracy in driving condition recognition and improves NVH evaluation efficiency
<u>Yıldırım, Ş., Yücekaya, A. D., Hekimoğlu, M., Ucal, M., Aydın, M. N., & Kalafat, İ. (2025). AI-driven predictive maintenance for workforce and service optimization in the automotive sector. Applied Sciences. (https://doi.org/10.3390/app15116282)</u>	Machine learning framework (Decision Tree, RF, LGBM, XGBoost)	Automotive after-sales service / CRM systems	Predictive maintenance & customer visit forecasting	XGBoost achieved highest accuracy; ML improved visit prediction and workforce optimization significantly

Paper	Method	Context	Focus / Topic	Key Findings
Mohammadian, N., Valilai, O. F. (2024). Influencer-driven paradigm shifts: Analyzing sustainability and customer behavior in cell phone and e-auto industries via social media analytics. IEEE Access. (https://doi.org/10.1109/ACCESS.2024.3449113)	System dynamics modeling + machine learning (sentiment analysis)	Automotive & consumer electronics (EVs, smartphones), social media (YouTube), omnichannel retail	Influencer impact on sustainability-oriented customer behavior	Influencers significantly shift sentiment from neutral/negative to positive, increasing engagement and sustainability awareness
Liu, Z., Liang, X., Li, L., Li, X., & Ou, W. (2024). Research on sustainable design of smart charging pile based on machine learning. Symmetry. (https://doi.org/10.3390/sym16121582)	Mixed-method approach (SAPAD + FAHP + FQFD + TBL + PSO-optimized Random Forest model)	Electric vehicle smart charging infrastructure design	Sustainable and user-driven charging pile design using AI and ML	Integrated framework improves alignment between user needs, design features, and sustainability factors through nonlinear predictive modeling
Mao, J., Hong, D., Ren, R., Li, X., Wang, J., & Nasr, E. S. A. (2020). Driving conditions of new energy logistics vehicles using big data technology. IEEE Access. (https://doi.org/10.1109/ACCESS.2020.3005529)	Empirical study + Big data analytics (K-means, Spark-based processing)	New energy logistics trucks in urban transportation systems	Driving condition analysis using big data for energy efficiency	Big data (Spark) improves analysis speed ~30× and reveals congestion dominates energy consumption and idle time

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Ballingall, S., Sarvi, M., & Sweatman, P. (2023). Standards relevant to automated driving system safety: A systematic assessment. Transportation Engineering. (https://doi.org/10.1016/j.treng.2023.100202)	Systematic assessment with thematic analysis of standards	Automated Driving Systems (ADS) safety assurance in automotive industry	Safety standards for ML-enabled automated driving systems	Current standards insufficient for ML-based ADS updates; lifecycle-wide safety cases, rigorous change control, and regulatory clarity needed for safe deployment
Esteller-Cucala, M., Fernandez, V., & Villuendas, D. (2020). Towards a data-driven culture in a Spanish automobile manufacturer: A case study. Journal of Industrial Engineering and Management. (https://doi.org/10.3926/jiem.3042)	Exploratory single case study	Automotive company digital platforms and decision-making processes	Data-driven organizational transformation and change management	Anticipating change risks supports transition to data-driven culture, but knowledge gaps and resistance hinder sustained adoption
Lopez-Vega, H., & Moodysson, J. (2023). Digital transformation of the automotive industry: An integrating framework to analyse technological novelty and breadth. Industry and Innovation. (https://doi.org/10.1080/13662716.2022.2151873)	Mixed-method approach (topic modelling + survey of inventors)	Automotive industry; autonomous vehicle patents (Sweden/Europe)	Digital transformation; technological novelty and breadth of AI/ML-driven innovations	Digital automotive technologies are mainly incremental and cross-sectoral; only few radical disruptive technologies exist, requiring incremental complements to realize full industry transformation.

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Redondo, R., Herrero, A., Corchado, E., & Sedano, J. (2020). A decision-making tool based on exploratory visualization for the automotive industry. Applied Sciences. (https://doi.org/10.3390/app10124355)	Mixed-method approach (ML-based exploratory visualization + case study)	Automotive manufacturing; waterjet cutting machines (predictive maintenance setting)	AI/ML-based decision support for failure detection and predictive maintenance	Extended HUEP visualization improves anomaly separation and supports decision-making; CMDS-based projections most effective for detecting machine failures.
Gabellini, M., Civolani, L., Calabrese, F., & Bortolini, M. (2024). A deep learning approach to predict supply chain delivery delay risk based on macroeconomic indicators: A case study in the automotive sector. Applied Sciences. (https://doi.org/10.3390/app14114688)	Case study + predictive analytics (LSTM, ARIMAX)	Automotive supply chain management	Supply chain delivery delay risk prediction	Deep learning LSTM with macroeconomic indicators improves delivery delay prediction accuracy over statistical benchmarks and univariate models.
Çıkmak, S. (2025). Pythagorean fuzzy AHP approach for evaluating the importance level of Industry 4.0 technologies in the automotive manufacturing industry. International Journal of Automotive Science and Technology. (https://doi.org/10.30939/ijastech.1522257)	Pythagorean fuzzy AHP (multi-criteria decision-making)	Automotive manufacturing (Türkiye Industry 4.0 adoption)	Evaluation of Industry 4.0 technology importance	Simulation & modeling, autonomous robots, and big data are top priorities; blockchain least important

Paper	Method	Context	Focus / Topic	Key Findings
Strazzullo, S., Cricelli, L., Troise, C., & Camilleri, M. A. (2025). Leveraging Industry 4.0 technologies for sustainable value chains: Raising awareness on digital transformation and responsible operations management. Sustainable Development. (https://doi.org/10.1002/sd.3211)	Inductive qualitative study with thematic analysis of expert interviews	Automotive supply chains and manufacturing firms	Digital transformation using Industry 4.0 for lean and sustainable operations	IoT, big data, AI, and simulation improve efficiency, reduce costs, and enable sustainable lean supply chains
Amro, A., Rehman, A. U., Ullah, I., & Anthony, B. O. (2025). Strategic role of customer-centric culture in driving engagement: Insights from Saudi Arabia's automotive sector. Cogent Business & Management. (https://doi.org/10.1080/23311975.2025.2567633)	Quantitative cross-sectional survey using PLS-SEM structural equation modeling	Automotive aftersales sector (Saudi Arabia)	Customer-centric culture and customer engagement dynamics	Customer-centric culture enhances engagement indirectly via customer experience; CX is strongest predictor
Gouveia, S., de la Iglesia, D. H., Abrantes, J. L., López Rivero, A. J., Silva, E., Gouveia, E., & Santos, V. (2024). Electrifying strategic management: Leveraging electric vehicles for sustainable value creation. Energies. (https://doi.org/10.3390/en17184581)	Systematic literature review with bibliometric analysis (PRISMA + VOSviewer; Scopus dataset)	Electric vehicle industry; strategic management, value creation, sustainability in digital automotive ecosystem	EV-driven strategic management, value creation, sustainability transitions	EV research has rapidly grown since 2019; three main clusters identified; key gaps exist in business and value-creation research; sustainability, policy, and circular economy are critical for EV ecosystem development

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Chehbi Gamoura, S., Damand, D., Lahrichi, Y., & Saikouk, T. (2025). FP-growth-based risk pattern discovery for dual cost-risk mitigation in resilient multi-sourcing order allocation under time-varying demand. International Journal of Production Economics. (https://doi.org/10.1016/j.ijpe.2025.109672)	Hybrid AI-driven optimization model (FP-Growth association rules + Risk-Mitigated Weighted Sum Method + simulation case study)	Automotive industry; multi-sourcing supply chain order allocation under time-varying demand	Risk-aware supplier selection, supply chain resilience, cost-risk optimization	FP-Growth uncovers hidden supplier risk patterns; hybrid model improves cost efficiency and resilience vs traditional methods; dynamic supplier exclusion enhances disruption mitigation
Mathai, J. (2024). Young consumers' perceptions of AI-powered in-car personalized services in a growing market for smart cars. Journal of Promotion Management. (https://doi.org/10.1080/10496491.2023.2251465)	Survey-based empirical study using SEM (Structural Equation Modeling)	AI-powered in-car personalized services in Indian smart cars (young consumers)	Technology adoption of AI-driven personalization in vehicles	Perceived usefulness and ease of use significantly increase behavioral intention; perceived risk weakens adoption relationships but is not directly significant
Hu, L., & Basiglio, A. (2023). A multiple-case study on the adoption of customer relationship management and big data analytics in the automotive industry. The TQM Journal. (https://doi.org/10.1108/TQM-05-2023-0137)	Multiple-case qualitative study using 18 semi-structured interviews, coding, and abductive analysis (NVivo)	Automotive industry firms in Italy and international OEM CRM/BDA ecosystems	CRM and big data analytics adoption for marketing and TQM digital transformation	CRM and BDA adoption improves customer-centric marketing, enhances TQM, and strengthens digital culture and business performance

Paper	Method	Context	Focus / Topic	Key Findings
Fodor, S., Ternai, K., Lengyel, E., & Gelei, A. (2025). Old problems in a new context: The Industry 4.0 readiness of automotive SMEs. Journal of Manufacturing Technology Management. (https://doi.org/10.1108/JMTM-03-2025-0249)	Mixed-method approach (abductive content analysis using BERTopic topic modeling + expert workshop + deductive coding with CMMI and SIMMI-based reference models)	Automotive SMEs (Industry 4.0 readiness assessment)	Industry 4.0 readiness model development (process + IT management capabilities)	I4.0 readiness depends on mature process and IT management; existing models overestimate readiness by mixing maturity and readiness.
Omeiza, D., Webb, H., Jirotko, M., & Kunze, L. (2022). Explanations in autonomous driving: A survey. IEEE Transactions on Intelligent Transportation Systems. (https://doi.org/10.1109/TITS.2021.3122865)	Literature review and taxonomy formulation.	Autonomous vehicle operations, regulatory standards, and stakeholder interactions.	Explainable artificial intelligence (XAI) and system transparency in autonomous driving.	Current AV explainability lacks user-centric design, suffers from low faithfulness, and requires standardized intermediate data access.

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<u>Salais-Fierro, T. E., Saucedo-Martinez, J. A., Rodriguez-Aguilar, R., & Vela-Haro, J. M. (2020). Demand prediction using a soft-computing approach: A case study of automotive industry. Applied Sciences. (https://doi.org/10.3390/app10030829)</u>	Hybrid soft-computing model combining Delphi method, fuzzy logic (Mamdani), and time-series neural networks (including NARX)	Automotive industry demand forecasting and supply chain planning	Demand prediction integrating expert judgement and historical data	Hybrid fuzzy-neural approach improves demand forecasting by capturing uncertainty and expert knowledge beyond traditional statistical methods
<u>Jiangwei, S., Chuan, Y., Yonggang, L., Shiquan, S., & Zheng, C. (2024). Battery fault diagnosis and anomaly detection based on data mining and big data analysis. Transactions of China Electrotechnical Society. (https://doi.org/10.19595/j.cnki.1000-6753.tces.231983)</u>	Statistical data mining and machine learning approach (Gaussian modeling, K-means clustering, Z-score, 3σ rule, t-SNE dimensionality reduction)	Electric scooter lithium-ion battery systems using cloud monitoring data	Fault diagnosis and cell-level anomaly detection in battery systems	Proposed hybrid statistical-ML framework enables accurate real-time detection and localization of system faults and abnormal battery cells using cloud data.

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Wang, Y., Zheng, R., Liang, C., & Shi, J. (2025). Inventory routing problem of automotive parts considering time-varying demands: A machine learning enhanced branch-and-price approach. Transportation Research Part E: Logistics and Transportation Review. (https://doi.org/10.1016/j.tre.2025.104297)	Mixed-method optimization (IP + machine learning–enhanced branch-and-price with online regression)	Automotive inbound logistics (multi-period auto parts supply for assembly plants)	Inventory routing optimization with time-varying demand	Machine learning–enhanced branch-and-price reduces computation time (>90%) and achieves near-optimal cost-efficient routing decisions
Raza, M., Jahangir, Z., Riaz, M. B., Saeed, M. J., & Sattar, M. A. (2025). Industrial applications of large language models. Scientific Reports. (https://doi.org/10.1038/s41598-025-98483-1)	Systematic literature review (multi-source screening of 2020–2024 papers + thematic analysis)	Cross-industry applications (automotive, healthcare, finance, education, e-commerce)	Large language models for industrial AI applications	LLMs improve automation, prediction, and decision-making across industries, including automotive predictive maintenance, despite ethical and computational challenges
Muth, M., Parsegyan, A. T., Litzinger, J., & Lingenfelder, M. (2025). Marketing predictions under macroeconomic volatility: Empirical evidence for automotive SMEs from a machine learning perspective. Journal of Marketing Analytics. (https://doi.org/10.1057/s41270-025-00436-0)	Empirical comparative machine learning study (21 forecasting methods incl. ML, econometric, ensembles)	Automotive SMEs; German car demand forecasting	Demand prediction under macroeconomic volatility	ML-integrated multivariate methods significantly outperform univariate and non-ML approaches in predictive accuracy for car registrations

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Rožanec, J. M., Kažič, B., Škrjanc, M., Fortuna, B., & Mladenčić, D. (2021). Automotive OEM demand forecasting: A comparative study of forecasting algorithms and strategies. Applied Sciences. (https://doi.org/10.3390/app11156787)	Mixed-method approach (CRISP-DM + agile), predictive analytics using 21 forecasting models (statistical, ML, streaming)	Automotive OEM supply chain, global engine components demand forecasting	Demand forecasting using ML and statistical algorithms	Global ML models outperform local ones; grouping products improves forecasting accuracy and error control
Mazur, M., Stopka, O., Stopková, M., Hanzl, J., Borucka, A., & Czerniak, R. (2025). Assessing the success of automotive sales transactions using selected machine learning algorithms. Applied Sciences. (https://doi.org/10.3390/app152111562)	Empirical study using machine learning classification models (RF, GBM, XGBoost, SVM, logistic regression)	Automotive CRM and premium vehicle sales (Mercedes-Benz, Polish market)	Sales success prediction and lead classification	Random Forest achieved best performance (AUC 0.90), enabling effective lead prioritization and resource allocation
Fu, W., Bian, H., Chen, J., & Jing, S. (2026). Tree-based machine learning intermittent demand forecasting for spare parts in electric vehicle manufacturing. World Electric Vehicle Journal. (https://doi.org/10.3390/wevj17030127)	Empirical study using tree-based machine learning (two-stage Random Forest, benchmark comparison)	Automotive aftermarket spare parts in EV-related manufacturing and service operations	Intermittent demand forecasting and inventory optimization	Two-stage Random Forest significantly outperforms statistical and ML benchmarks, improving forecasting accuracy and inventory decisions

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El Kihel, Y., Zouggar Amrani, A., Ducq, Y., Amegouz, D., & Lfakir, A. (2023). Methodology combining Industry 4.0 technologies and KPI reliability for supply chain performance. International Journal of Computer Integrated Manufacturing. (https://doi.org/10.1080/0951192X.2022.2162605)	Mixed-method framework (GRAILOG + PPTechIP cross-analysis + case study at PSA)	Automotive supply chain (PSA, Industry 4.0 transformation)	Industry 4.0 technologies impact on KPI reliability and supply chain performance	Integrating Industry 4.0 technologies improves KPI reliability; radar-based PP identifies digital transformation gaps and investment priorities
Poushneh, A., & Vasquez-Parraga, A. (2024). The effect of semiautonomous augmented reality on customer experience and augmentation experience. European Journal of Marketing. (https://doi.org/10.1108/EJM-05-2022-0402)	Mixed-method empirical study (preliminary survey, field study in car dealership, lab experiment)	Automotive retail/service (car dealerships using augmented reality systems)	Semiautonomous AR, customer experience, perceived control, technology adoption	Perceived control increases augmentation experience; enriched experience and satisfaction mediate effects; continuation intention not significantly different between AR and non-AR users
Sathyan, R., Parthiban, P., Dhanalakshmi, R., & Minz, A. (2021). A combined big data analytics and fuzzy DEMATEL technique to improve the responsiveness of automotive supply chains. Journal of Ambient Intelligence and Humanized Computing. (https://doi.org/10.1007/s12652-020-02524-8)	Mixed-method (big data analytics + sentiment analysis + ANN forecasting + fuzzy DEMATEL expert evaluation)	Automotive industry (Indian mid-SUV vehicle, supply chain responsiveness)	Demand forecasting from online reviews and vehicle attribute prioritization	Neural network achieved 95.5% accuracy; price and safety features most influential purchase attributes

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Kaiser, C., Stocker, A., Viscusi, G., Fellmann, M., & Richter, A. (2021). Conceptualising value creation in data-driven services: The case of vehicle data. International Journal of Information Management. (https://doi.org/10.1016/j.ijinfomgt.2021.102335)	Design science research; expert interviews; case-based evaluation	Automotive industry; vehicle data ecosystems; connected mobility services	Data-driven value creation; multi-actor ecosystem modeling	Identifies key actors and data-sharing relationships enabling vehicle data-driven services; proposes validated conceptual ecosystem model
Piepoli, A., Arcidiacono, F., Basile, L. J., Pellegrino, R., Schupp, F., & Zuehlke, T. (2024). The interplay between Industry 4.0 technologies and business performance: Evidence from a multiple case study in the automotive sector. IEEE Engineering Management Review. (https://doi.org/10.1109/EMR.2023.3328780)	Multiple case study	Automotive supply chains; automotive suppliers	Industry 4.0 adoption; organizational strategy; business performance	Planning and external collaboration are critical; advanced I4.0 improves business performance more strongly
Chomachaei, F. (2025). The effect of ride-hailing platforms on the automobile industry in the United States: An empirical study. IEEE Transactions on Engineering Management. (https://doi.org/10.1109/TEM.2025.3605820)	Empirical study; panel data regression (fixed effects)	Automotive industry; ride-hailing platforms (U.S. manufacturers)	Ride-hailing impact on automotive financial performance	Ride-hailing growth increases manufacturers' financial performance via higher production and stimulates vehicle demand

Paper	Method	Context	Focus / Topic	Key Findings
Varriale, V., Michelino, F., & Godinho Filho, M. (2025). R&D processes with artificial intelligence: Mapping applications and impacts. IEEE Transactions on Engineering Management. (https://doi.org/10.1109/TEM.2025.3592217)	Multiple case study; empirical analysis; panel data econometric model; clustering/content analysis of literature practices	Automotive manufacturing; automotive supply chains; ride-hailing mobility ecosystem; R&D and engineering processes across industries	Industry 4.0 adoption, ride-hailing impact, AI in R&DE, digital transformation and value creation in automotive sector	Industry 4.0 adoption improves supplier performance; ride-hailing increases vehicle demand and manufacturer profitability; AI enhances R&DE efficiency, innovation, and reduces cost/time across automotive and related industries
Kiatcharoenpol, T., & Sirisawat, P. (2026). Evaluating critical barriers to Industry 4.0 adoption in the Thai automotive sector using an integrated fuzzy BWM-PROMETHEE II-DEMATEL framework. IEEE Access. (https://doi.org/10.1109/ACCESS.2026.3669303)	Integrated fuzzy MCDM framework (Fuzzy BWM + PROMETHEE II + DEMATEL); expert-based evaluation	Thai automotive manufacturing sector; Industry 4.0 adoption in supply chains	Industry 4.0 barriers assessment, prioritization, and causal analysis under uncertainty	High initial investment, supply chain integration, and interoperability are key causal barriers; scalability is major effect barrier; financial and ecosystem readiness are critical for I4.0 adoption success

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Sieberg, P. M., Blume, S., Reicherts, S., Maas, N., & Schramm, D. (2022). Hybrid state estimation: A contribution towards reliability enhancement of artificial neural network estimators. IEEE Transactions on Intelligent Transportation Systems. (https://doi.org/10.1109/TITS.2021.3055800)	Hybrid estimation model (ANN + physics-based model) using Unscented Kalman Filter; simulation-based evaluation	Automotive vehicle dynamics (passenger cars); sensor-based state estimation	Reliability enhancement of AI/ML estimators; hybrid data-driven + model-based estimation; sensor fault robustness	Combining neural networks with physics-based models via Kalman filtering improves estimation reliability and accuracy, especially under sensor faults and missing data scenarios
Bodendorf, F., & Franke, J. (2024). The technological transformation process for dynamic capabilities in business operations. IEEE Transactions on Engineering Management. (https://doi.org/10.1109/TEM.2024.3349478)	System dynamics simulation study using empirical automotive data and multi-model causal analysis	Automotive industry (production systems, digital transformation)	Technological transformation, digital maturity, dynamic capabilities, competitiveness	Digital maturity and technological transformation strongly enhance dynamic capabilities and competitiveness via feedback-driven adaptation processes
Nayal, K., Raut, R. D., Queiroz, M. M., & Priyadarshinee, P. (2024). Digital supply chain capabilities: Mitigating disruptions and leveraging competitive advantage under COVID-19. IEEE Transactions on Engineering Management. (https://doi.org/10.1109/TEM.2023.3266151)	Empirical survey with structural equation modeling	Automotive supply chain (Indian SME OEM sector)	Digital supply chain capabilities, resilience, agility, sustainability, competitive advantage	Digital supply chain capabilities improve resilience and agility, enabling sustainable practices that enhance competitive advantage

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Tan, J., Luan, T. H., Guan, W., Wang, Y., Peng, H., Zhang, Y., Zhao, D., & Lu, N. (2024). Beam alignment in mmWave V2X communications: A survey. IEEE Communications Surveys & Tutorials. (https://doi.org/10.1109/COMST.2024.3383093)	Literature review (survey of beam alignment techniques and performance comparison)	Automotive V2X communications using mmWave networks for autonomous vehicles	Beam alignment in mmWave V2X, communication reliability, autonomous driving connectivity	Efficient beam alignment is critical for reliable mmWave V2X, with trade-offs across methods under blockage and latency constraints
Teusch, J., Gremmel, J. N., Koetsier, C., Johora, F. T., Sester, M., Woisetschläger, D. M., & Müller, J. P. (2023). A systematic literature review on machine learning in shared mobility. IEEE Open Journal of Intelligent Transportation Systems. (https://doi.org/10.1109/OJITS.2023.3334393)	Systematic literature review (PRISMA-based) with structured database screening and qualitative synthesis of ML studies	Shared mobility systems (ride-sharing, bike-sharing, e-scooters, car-sharing, urban transport services)	Machine learning for decision support in shared mobility operations and management	ML enhances demand prediction, dispatching, pricing, and planning; key gaps include interpretability, data limitations, and e-scooter research
Garikapati, D., & Shetiya, S. S. (2024). Autonomous vehicles: Evolution of artificial intelligence and the current industry landscape. Big Data and Cognitive Computing. (https://doi.org/10.3390/bdcc8040042)	Mixed-method approach (semi-structured expert interviews, qualitative analysis)	Automotive industry innovation, AI-enabled vehicles and manufacturing	AI-driven innovation, autonomous driving, and digital transformation	AI improves safety, efficiency, and innovation but faces challenges in ethics, regulation, data quality, and talent shortages

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Turienzo, J., Cabanelas, P., & Lampón, J. F. (2023). Business models in times of disruption: The connected and autonomous vehicles domino effect. Journal of Business Research. (https://doi.org/10.1016/j.jbusres.2022.113481)	Literature review + quantitative publication trend analysis + conceptual framework	Autonomous driving and automotive AI systems	Evolution of AI/ML/DL/GenAI in autonomous vehicles	AI use in autonomous vehicles is rapidly increasing, with deep learning dominating perception and decision-making systems while higher autonomy increases system complexity and software demands
Rao, Sriganesh k. (2021). data-driven business model innovation for 6g. journal of business research. (https://doi.org/10.13052/jicts2245-800x.935)	Exploratory qualitative study using Grounded Theory interviews	Mobility industry (connected and autonomous vehicles ecosystem)	Business model innovation under autonomous and connected vehicle disruption	Data-driven ecosystems, alliances, and servitization (B2B/MaaS) are key; value creation depends on data management, regulation, and partnerships

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Kiran, b. r., sobh, i., talpaert, v., mannion, p., al sallab, a. a., yogamani, s., & pérez, p. (2021). deep reinforcement learning for autonomous driving: a survey. journal of ict standardization. (https://doi.org/10.1109/tits.2021.3054625)	Conceptual framework and literature-based analysis	6G-enabled smart cities, intelligent transport systems, Industry 4.0, healthcare, digital ecosystems	Data-driven business model innovation leveraging real-time and dark data using AI analytics	6G drives data explosion; AI-driven data models enable sustainability and competitiveness; traditional analytics platforms are insufficient
chai, t. y., & nizam, i. (2021). impact of artificial intelligence in automotive industries transformation. international journal of production research. (https://doi.org/10.24924/ijise/2021.04/v9.iss2/01.35)	Survey + theoretical analysis of DRL algorithms and autonomous driving architectures	Autonomous driving systems and intelligent transportation	Deep reinforcement learning for autonomous driving decision-making	DRL enables end-to-end driving policies but faces safety, scalability, and sim-to-real transfer challenges
llopis-albert, c., rubio, f., & valero, f. (2021). impact of digital transformation on the automotive industry. applied system innovation. (https://doi.org/10.1016/j.techfore.2020.120343)	Empirical quantitative survey (questionnaire + regression analysis)	Automotive industry transformation (manufacturing, autonomous vehicles, marketing, smart factories)	Impact of AI on automotive industry transformation	AI significantly improves leadership, autonomous vehicles, smart factories, and marketing performance in automotive sector

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shrimal, h. (2024). integration of ai-powered vehicles with smart city infrastructure to transform the future of automotive world. technological forecasting and social change. (https://doi.org/10.4271/2024-28-0028)	Conceptual framework / technical analysis	Smart cities and autonomous vehicle ecosystems	Integration of AI-powered vehicles with smart city infrastructure	Integration enables safer, more efficient, and sustainable urban mobility through data-driven traffic and infrastructure management
markov, k., & vitliemov, p. (2020). logistics 4.0 and supply chain 4.0 in the automotive industry. international journal of lean six sigma. (https://doi.org/10.1088/1757-899x/878/1/012047)	Expert survey	Automotive logistics and supply chain operations	Implementation of Logistics 4.0 and Supply Chain 4.0 technologies	Production visibility and blockchain improve logistics performance, while cybersecurity and data quality remain key barriers
borg, m., englund, c., wnuk, k., duran, b., levandowski, c., gao, s., tan, y., kaijser, h., lönn, h., & törnqvist, j. (2020). safely entering the deep: a review of verification and validation for machine learning and a challenge elicitation in the automotive industry. ieee transactions on industrial informatics. (https://doi.org/10.2991/jase.d.190131.001)	Systematic literature review and industry survey	Autonomous driving and safety-critical automotive systems	Verification and validation of machine learning safety	Existing safety standards poorly fit ML systems; simulation and safety architectures recommended for reliable certification
gupta, s., amaba, b., mcmahon, m., & gupta, k. (2021). the evolution of artificial intelligence in the automotive industry. production and operations management. (https://doi.org/10.1109/rams48097.2021.9605795)	Conceptual framework / industry analysis	Automotive supply chain and vehicle lifecycle management	Reliability and maintainability transformation through AI and big data	AI-driven analytics enhances vehicle reliability, operational performance, and customer experience across the automotive ecosystem

Paper	Method	Context	Focus / Topic	Key Findings
fernandes, j., reis, j., melão, n., teixeira, l., & amorim, m. (2021). the role of industry 4.0 and bpmn in the arise of condition-based and predictive maintenance: a case study in the automotive industry. socio-economic planning sciences. (https://doi.org/10.3390/app11083438)	Case study	Automotive manufacturing operations	Condition-based and predictive maintenance implementation	Sensor-based monitoring enabled early fault detection, reduced downtime, and generated significant cost savings and productivity improvements